

Great Financial Crisis and Dollar Dominance

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Abstract

The majority of international trade transactions are invoiced in US dollars, which has key implications for the global macroeconomy. Understanding the forces that sustain this equilibrium is therefore crucial. This paper studies how temporary shocks to US dollar financing availability can generate persistent shifts in firms invoicing decisions in international trade and its aggregate implications. Using the universe of Chilean export and import transactions from 2004 to 2019, we show that after the Global Financial Crisis (GFC), the share of imports invoiced in US dollars decreased by nearly 10 percentage points, with a corresponding and permanent rise in the probability of Chilean peso invoicing. We develop a model in which importers face complementarities between the currency used for trade invoicing and that used for financing imported inputs. UIP deviations alter borrowing costs and thus invoicing choices for trade. Consistent with the model's predictions, we find that a temporary spike in US dollar denominated financing cost led firms to substitute the US dollar with the domestic currency when invoicing imports, explaining half of the aggregate dynamics. The effect is stronger for highly dollar-exposed importers and weaker in markets where competitors predominantly invoice in US dollars due to strategic complementarities in invoicing. At the aggregate level, the shift in invoicing patterns reduced the sensitivity of import inflation to exchange rate fluctuations while increasing that of the terms of trade.

JEL Codes: F14, F31, F33, F41.

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The global economy is dominated by the US Dollar, which serves as the dominant currency for invoicing international trade, pricing global assets, and holding reserves (Gopinath, 2015; Gopinath and Stein, 2021). Existing research explains this dominance through the supply of safe assets, complementarities between financing and invoicing decisions, and the role of the US Dollar as a vehicle currency (Farhi and Maggiori, 2018; Devereux and Shi, 2013). Yet, we know little about the shocks that could temporarily or permanently weaken this dominance. This paper examines how disruptions in US Dollar funding and deviations from uncovered interest parity (UIP) can produce persistent effects on firms invoicing behavior in international trade. While past shifts in monetary dominance followed structural changes, we show that temporary shocks, such as those during the Global Financial Crisis (GFC), can also generate lasting consequences in trade invoicing patterns (Amiti and Weinstein, 2011; Ahn et al., 2011; Salomao and Varela, 2022; Kalemli-Ozcan and Varela, 2021).

Using the universe of export and import transactions from Chile between 2004 and 2019, we show that, starting during the GFC, the share of imports invoiced in US dollars fell by nearly 10 percentage points, with a corresponding and permanent rise in the share of Chilean peso invoicing. Using an Olley-Pakes decomposition of the observed change in aggregate US dollar invoicing (Melitz and Polanec, 2015), we show that the change in invoicing is driven by incumbent importers that, after the GFC, started invoicing the same good from the same origin in domestic currency rather than in US dollar. The switch occurs at the firm-origin-product level and appears permanent, as firms rarely revert once they switch.

To rationalize these empirical patterns, we build on Bahaj and Reis (2020). In our model, imported inputs play the role of working capital, which require financing in advance. Importers face within-firm complementarities between the currency used for trade invoicing and that used for financing imported inputs. Differentials in financing costs denominated in different currencies, which can be proxied by UIP deviations, affect the invoicing choice through their impact on the optimal borrowing currency. As in Crowley et al. (2020), we introduce currency- and firm-specific fixed costs of invoicing. These costs decline as more firms use the same currency, capturing strategic complementarities in invoicing. Moreover, we assume that fixed costs include a sunk-cost component, such as the setup cost of managing transactions in a new currency: once incurred, the ongoing cost of using that currency remains permanently lower.

The model generates three key testable implications. First, when US dollar borrowing cost rises, firms become more likely to substitute the US dollar with the domestic currency. Second, conditional on rising borrowing costs in US dollar, firms with a higher share of import invoiced in US dollar firms should display a greater propensity to switch to domestic-currency invoicing. Third, this effect should weaken when a larger share of competitors invoice in US dollar, reflecting stronger strategic complementarities in invoicing.

We test and verify the key predictions of our framework using import and export transactions from Chile and leveraging both cross-sectional variation and an event-study design. During normal times, borrowing in US dollar is typically cheaper than in domestic currency,

but during the GFC, the scarcity of dollar liquidity reversed this relationship, making peso borrowing more attractive. We show that UIP deviations is a key predictor of invoicing choices, explaining half of the aggregate increase in domestic currency invoicing. Moreover, consistently with our framework, importers with higher pre-GFC exposure to US dollar imports were more likely to switch to the Chilean peso when dollar financing became relatively more expensive at the onset of the GFC. Conversely, firms more exposed to competitors invoicing in US dollar were less likely to switch, consistent with strategic complementarities in invoicing across importers. Lastly, simulating a temporary scarcity of dollar liquidity, similar to that observed during the GFC, generates long-lasting effects on firms invoicing decisions in the model. The presence of sunk costs implies that temporary shocks can permanently alter the relative attractiveness of different invoicing currencies. Consistent with this mechanism, we find strong empirical evidence of sunk costs in invoicing choices: past invoicing decisions robustly predict current ones, supporting the models hysteresis mechanism in currency use.

To conclude, we examine the macroeconomic implications of the decline in the aggregate share of dollar-denominated imports, particularly its effects on exchange rate pass-through, the dynamics and sensitivity of the terms of trade and import inflation. Consistent with previous literature, invoicing plays a crucial role in determining the degree of exchange rate pass-through, which tends to be higher when transactions are invoiced in a dominant currency and lower when invoiced in a local currency (Gopinath et al., 2010; Barbiero, 2021). As a result, by 2019, the larger share of imports invoiced in local currency had reduced the sensitivity of import inflation to exchange rate fluctuations by 5% compared to 2007, while increasing the sensitivity of the terms of trade.

Related Literature Most models of currency dominance have in common the fact that they either define under which conditions a currency becomes dominant or they define under which conditions a currency remains dominant (Eichengreen et al., 2018; Chahrour and Valchev, 2022; Farhi and Maggiori, 2018; Gopinath and Stein, 2021).¹ Fewer papers instead focus on forces and shocks that can perturb the current dollar dominant equilibrium. Chahrour and Valchev (2023) and Corsetti et al. (2022) share the same perspective in analyzing how geo-political fragmentation and Brexit, respectively, affect the current dominant currency equilibrium. Bahaj and Reis (2020) and Bahaj and Reis (2022) focus on currency competition in the central banks' swap line market as a channel through which the People's Bank of China jump-started the international use of the renminbi, challenging the dominant role of the dollar. Our paper complements this literature concerned with discontinuities in the international monetary system as we shed light on how large macroeconomic events such as the Great Financial Crisis might impair the dominance of a single currency and the mechanism that could lead to the rise of another competitive currency.²

¹Other recent contributions such as Gopinath et al. (2020) and Mukhin (2022) focus on understanding the consequences that the US dollar dominance has on international trade dynamics, and the global economy in general. Refer to Gourinchas et al. (2019) for a survey.

²Only few other papers share the same approach and focus on a large aggregate transition in invoicing patterns like in our setting: Garofalo et al. (2024), which focuses on the UK after Brexit, and Mehl and Mlikota

Our theoretical framework emphasizes the crucial role of trade finance. International trade transactions require external financing, which is typically heavily dollarized and sourced locally via domestic banks (Niepmann and Schmidt-Eisenlohr, 2017). The forces driving currency dominance in trade and finance are closely intertwined. The availability of safe, liquid, short-term government-backed liabilities fosters invoicing dominance, as shown by Coppola et al. (2023) and Eren and Malamud (2022). Similarly, Chahrour and Valchev (2023) highlights that dollar dominance arises from the need for both parties in an international trade transaction to post collateral. Gopinath and Stein (2021) proposes a feedback loop between invoicing and financing choices, showing that UIP deviations are necessary to the dominance of a currency. While their contribution is theoretical, we empirically document how shocks to dollar financing, proxied by UIP shocks, can affect invoicing decisions, in line with their predictions.^{3 4}

Moreover, Kalemli-Ozcan and Varela (2021) and Salomao and Varela (2022) empirically document how UIP deviations influence foreign investment and the currency of invoicing for investment financing, respectively. We complement these contributions by providing evidence that the scarcity of dollar-denominated trade finance – or, equivalently, large UIP deviations – during the GFC not only affected trade volumes (Bruno and Shin, 2019; Amiti and Weinstein, 2011), but also impacted invoicing decisions in international trade flows, with potential long-term consequences for dollar dominance.

Extensive literature focus on the firm-level determinants of invoicing choices and heterogeneous pass-through rate using micro-level data. Several papers have focused on the importance of variable markups and strategic complementarities in price-setting (Amiti et al., 2019, 2022), international input intensity and operational hedging (Amiti et al., 2014), firm characteristics such as size and productivity (Berman et al., 2012).⁵ We show that, on one hand, financial considerations are a key determinant of invoicing choices. On the other hand, standard firm-level margins are crucial not only for understanding cross-sectional differences in invoicing patterns but also for explaining the adoption of different currencies.

The paper is organized as follow. Sections 1 and 2 describe the data and invoicing dynamics in Chile, respectively. Section 3 presents the theoretical framework. Section 4 provides evidence consistent with the key theoretical mechanisms. Section 5 assesses the aggregate implication of the observed changing invoicing patterns.

(2023) on the EU enlargement.

³Differently from us Farhi and Maggiori (2018) explores the impact of invoicing on safe asset determination, while we focus on the same link but in the opposite direction.

⁴In our analysis, we abstract from the possibility of foreign exchange hedging (Lyonnet et al., 2022). Using data from Chile, Alfaro et al. (2021) documents the development and increased depth of the hedging market in the country starting in the early 2000s. While hedging is a relevant option for managing exchange rate risk when invoicing in foreign currency, our work is complementary in focusing on the role of dollar borrowing, which is generally cheaper on average.

⁵See Burstein and Gopinath (2014), Itskhoki (2021), and Gopinath and Itskhoki (2022) for the most recent survey articles on this vast literature.

1 Data

Customs and Invoicing Data The primary data source is the universe of Chilean trade transactions from the Chilean Custom Agency (Aduanas). The agency provides transaction-level data for each firm, product, destination/origin, currency, and day, covering both export and import flows. The dataset includes the quantity exchanged (in kilograms and the product's unit of measure), the 8-digit HS classification of goods, and the transaction value (FOB) in US dollars. Crucially, it also records the invoicing currency, covering major currencies (US dollar, Euro, Chilean peso, UK pound, and Chinese yuan) as well as regional Latin American currencies. Additionally, an anonymous firm identifier allows for constructing a panel of Chilean importers and exporters. Import transaction data are available from 2007 to 2019, while export transaction data cover the period from 2009 to 2019. Since our main interest is the impact of the Great Financial Crisis on the use of the US dollar in international trade, we supplement the Chilean Customs Agency dataset with a similar dataset from [Garcia-Marin et al. \(2019\)](#).⁶ This additional dataset contains the same information as the official customs records starting from 2004, allowing us to capture the pre-GFC period and rule out potential pre-trends. The dataset covers customs transactions from Chile's main trading partners. It accounts for 98% of total Chilean trade in value and quantity compared to the official data, which give us confidence in the quality and the validity of the data.

UIP Data To proxy for domestic and foreign borrowing costs in constructing UIP deviations between the U.S. dollar and the Chilean peso, we use publicly available data from Chile's financial regulatory authority, the *Comisión para el Mercado Financiero* (CMF), which collects and reports aggregate data on the banking sector. They provide average deposit and lending rates at monthly frequency, distinguishing between domestic and foreign currency-denominated loans, as well as variations by maturity, loan size, and client type (commercial vs. non-commercial). We focus on commercial lending rates for short-term (one year) loans denominated in Chilean pesos and US dollars. Short-term loan rates are particularly relevant, as trade financing for international transactions heavily depends on this type of credit ([Amiti and Weinstein, 2011](#); [Ahn et al., 2011](#); [Schmidt-Eisenlohr, 2013](#)). The data cover the period around the GFC but are only available until 2014, when regulatory changes disrupted consistent comparisons between domestic and US dollar-denominated loans. As a result, we can document financing cost dynamics primarily between 2007 and 2009, along with adjacent periods.

We complement the data on borrowing costs with several measures of the bilateral exchange rate between the U.S. dollar and the Chilean peso. We use both the spot exchange rate and the realized one-year-ahead exchange rate to construct our measure of realized deviations from UIP. Daily spot U.S. dollar-peso exchange rates are obtained from Datastream. To construct the UIP premium, we rely on monthly one-year-ahead exchange rate forecasts from Bloomberg. The dataset begins in June 2006 and includes forecasts of the U.S. dollar-

⁶We kindly thank Santiago Justel for sharing the dataset with us.

peso exchange rate from approximately 160 financial institutions and market makers. We aggregate these forecasts by taking their cross-sectional average at the quarterly frequency.⁷ Finally, we use non-deliverable forward (NDF) outright contracts for the U.S. dollar-peso from Bloomberg to construct measures of covered interest parity.

Lastly, we construct measures of uncovered interest parity deviations relative to the US dollar for a panel of emerging market currencies from 2006 to 2012. We include Brazil, Colombia, Mexico, Peru, South Africa, Turkey, Hungary, Poland, Israel, and Indonesia. Daily spot and one-month ahead exchange rates are sourced from Datastream and Thomson Reuters, with all rates defined against the US dollar. To compute the one-month interest rate, we take the difference between the logarithm of the one-month ahead exchange rate and the logarithm of the spot exchange rate, assuming covered interest parity holds. We then calculate weekly averages for spot exchange rates, one-month interest rate differentials, and deviations from uncovered interest parity. For a subset of these countries, we collect information on local currency invoicing in imports from [Boz et al. \(2020\)](#) for the period 2004-2015. We use this information in a cross-country analysis in Appendix [A.7](#).

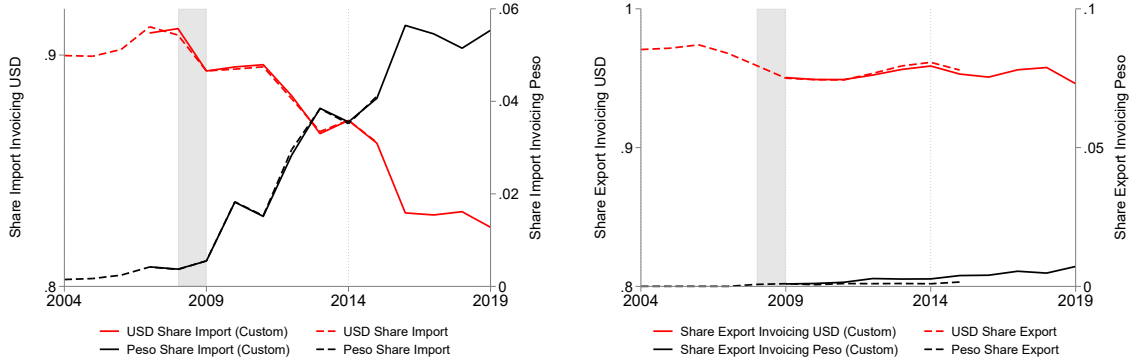
2 Macro and Micro Dynamics in Invoicing

In this section, we use firm-level customs data to document three main empirical facts about invoicing dynamics in Chilean trade. First, we show that, following the GFC, the share of aggregate imports invoiced in U.S. dollars declined, while invoicing in Chilean peso increased, with no corresponding changes on the export side. Second, using a decomposition in the spirit of [Melitz and Polanec \(2015\)](#), we find that these shifts were primarily driven by incumbent importers, rather than by entry or reallocation across importers, which reduced their reliance on the U.S. dollar and increased invoicing in local currency. Third, we show that both the probability of invoicing imports in local currency and the probability of switching from U.S. dollar to local-currency invoicing permanently increased after the GFC.

Aggregate. Following the GFC, the share of imports invoiced in U.S. dollars declined by 10 percentage points, while no change in invoicing patterns occurred on the export side. Figure [1](#) shows that, at the onset of the GFC, the share of imports denominated in U.S. dollars was approximately 90 percent. Over the following ten years, it declined to just above 80 percent. At the same time, the share of imports invoiced in Chilean peso increased from virtually 0 percent in 2004 to approximately 6 percent in 2019. Notably, we do not observe similar changes in the share of exports invoiced in different currencies: the shares of exports denominated in the main currencies (U.S. dollar, Euro, and Chilean peso) remain stable over time.

⁷Table [8](#) in Appendix [A](#) reports summary statistics for the exchange rate forecasts.

Figure 1: Aggregate Invoicing Shares for Imports and Exports



Note: The Figure shows the time series of the aggregate share of imports denominated in US dollars and Chilean pesos for Chile between 2004-2019. The left vertical axis of Figure 1 reports the share of Chilean imports denominated in US dollar. The right axis shows the share of Chilean imports denominated in Chilean pesos. Data from 2007 (2009) for imports (exports) are from Chilean Customs Agency. Data before 2007 (2009) are from [Garcia-Marin et al. \(2019\)](#). The grey shaded area represents the NBER recession period.

Firm Level Decomposition. To understand the margins driving the dynamics of the aggregate invoicing shares reported in Figure 1, we perform a dynamic Olley-Pakes decomposition in the spirit of [Melitz and Polanec \(2015\)](#). The goal of this section is to quantify the relative role played by: (i) firms' entry and exit and the reallocation of imports between and within firms; (ii) firms' sourcing strategies at the origin $\times$ product level; and (iii) invoicing currency adjustment margins.

Let $y_{\eta,t}^k$ denote imports of unit η invoiced in currency k at time t , and $y_{\eta,t} \equiv \sum_k y_{\eta,t}^k$. Thus, we can define $\lambda_{\eta,t}^k$ as the share of η 's imports invoiced in currency k , $\lambda_{\eta,t}^k \equiv \frac{y_{\eta,t}^k}{\sum_k y_{\eta,t}^k} = \frac{y_{\eta,t}^k}{y_{\eta,t}}$. Let γ index the aggregate containing η , $y_{\gamma,t} \equiv \sum_{\eta \in \gamma} y_{\eta,t}$. Thus, we can define η 's share of total imports at level γ as $s_{\gamma,t}^\eta \equiv \frac{y_{\eta,t}}{y_{\gamma,t}}$. For a given variable x , define $\bar{x}_t \equiv \frac{1}{2}(x_t + x_{t-1})$ and $\Delta x_t \equiv x_t - x_{t-1}$. Throughout this section, we use i to denote importers and m to denote origin-product pairs, with the product defined at the HS8 level.

Thus, the change in the share of total imports invoiced in currency k at time t , $\Delta \lambda_t^k$, can be decomposed as follow:

$$\Delta \lambda_t^k = \underbrace{\sum_{i \in I_t} \Delta s_t^i \bar{\lambda}_{i,t}^k}_{\text{Incumbent - Between}} + \underbrace{\sum_{i \in I_t} \Delta \lambda_{i,t}^k \bar{s}_t^i}_{\text{Incumbent - Within}} + \underbrace{\sum_{i \in I_t^+} \lambda_{i,t}^k s_t^i - \sum_{i \in I_t^-} \lambda_{i,t-1}^k s_{t-1}^i}_{\text{Net Entry}}, \quad (1)$$

$$\Delta \lambda_{i,t}^k = \underbrace{\sum_{m \in I_{it}} \Delta \lambda_{im,t}^k \bar{s}_{i,t}^m}_{\text{Within Origin-Product}} + \underbrace{\sum_{m \in I_{it}} \bar{\lambda}_{im,t}^k \Delta s_{i,t}^m}_{\text{Between Origin-Product}} + \underbrace{\sum_{m \in I_{it}^+} \lambda_{im,t}^k s_{i,t}^m - \sum_{m \in I_{it}^-} \lambda_{im,t-1}^k s_{i,t-1}^m}_{\text{Net Entry Origin-Product}} \quad \forall i \in I_t, \quad (2)$$

$$\Delta \lambda_{im,t}^k = \underbrace{\lambda_{im,t}^k \mathbf{1}_{\{A_{im,t}^k\}} - \lambda_{im,t-1}^k \mathbf{1}_{\{B_{im,t-1}^k\}}}_{\text{Currency Net Entry}} + \underbrace{\Delta \lambda_{im,t}^k \mathbf{1}_{\{C_{im,t}^k\}}}_{\text{Within-Currency}} \quad \forall i \in I_t, \forall m \in I_{i,t}, \quad (3)$$

where I_t is the set of continuing importers, while I_t^- and I_t^+ denote the sets of exiting and entering importers; similarly, I_{it} , I_{it}^- , and I_{it}^+ denote the sets of origin-product pairs m from which importer i imports in both t and $t-1$, only in $t-1$, and only in t , respectively.

The first layer of the decomposition, Equation (1), allows us to assess whether the change in the share of aggregate imports invoiced in currency k is driven by net entry of importers or incumbent importers, disentangling the role played by changes in incumbent market shares from changes in their invoicing choices (within-firm dynamics versus between-firm reallocation). The second layer, Equation (2), assesses whether the observed change in incumbents' invoicing shares is driven by shifts in sourcing across origin-product pairs. Finally, at the most granular level, Equation (3) shows that the share of imports invoiced in currency k from origin-product pair m can be decomposed into: (i) the currency k that is adopted in t but was not used in the preceding period $t - 1$, indicating new adoption ($1_{\{A_{im,t}^k\}}$); (ii) the currency k that is dropped in t after having been used in $t - 1$ ($1_{\{B_{im,t-1}^k\}}$); and (iii) the change in the use of currency k when the importer continues using it across both periods ($1_{\{C_{im,t}^k\}}$).

Table 1: Olley-Pakes Decomposition of Imports Invoicing

	US dollar	Chilean peso	Euro	Others
Δ Aggregate Invoicing Share	-9.452	6.093	3.611	-0.251
Net Entry	-1.769 (18.7%)	0.778 (12.8%)	0.813 (22.5%)	0.178 (29.3%)
Incumbents	-7.683 (81.3%)	5.315 (87.2%)	2.798 (77.5%)	-0.430 (70.7%)
Incumbents	-7.683	5.315	2.798	-0.430
Within Firm	-4.560 (59.4%)	6.367 (85.8%)	-1.337 (24.4%)	-0.470 (92.0%)
Between Firm	-3.123 (40.6%)	-1.052 (14.2%)	4.134 (75.6%)	0.041 (8.0%)
Within Firm	-4.560	6.367	-1.337	-0.470
Net Entry Origin-Product	0.975 (15.0%)	2.131 (33.5%)	-2.980 (55.7%)	-0.126 (17.2%)
Within Origin-Product	-5.139 (78.9%)	3.607 (56.6%)	2.007 (37.5%)	-0.475 (65.0%)
Between Origin-Product	-0.396 (6.1%)	0.630 (9.9%)	-0.363 (6.8%)	0.130 (17.8%)
Within Origin-Product	-5.139	3.607	2.007	-0.475
Currency Net Entry	-2.170 (42.2%)	2.060 (57.1%)	1.903 (94.8%)	-0.505 (94.2%)
Within Currency	-2.968 (57.8%)	1.546 (42.9%)	0.104 (5.2%)	0.031 (5.8%)

Notes: The Table reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin-product level for Chilean imports between 2007 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. We report the cumulative decomposition. Numbers are in percentage. Numbers in parenthesis account for how much each margin explains in a given layer. Data are from the Chilean Customs Agency.

Table 1 reports the decomposition of the cumulative change in invoicing shares in Chilean imports between 2007 and 2019 across the four invoicing currencies: US dollar, Chilean peso, Euro, and the Other category grouping all remaining currencies. More than 80 percent of the cumulative decline in the aggregate US dollar share is explained by incumbent firms ($-7.683 / -9.452 \approx 0.81$), with the remainder due to net entry. Among incumbents, nearly 60 percent of the contribution ($-4.560 / -7.683 \approx 0.59$) comes from the within-firm component, with the rest due to between-firm reallocation. The main driver is not that US dollar-invoicing importers are losing market share, but rather that they invoice less frequently in US dollars. Similarly, the increased use of the Chilean peso is almost entirely driven by incumbents that start invoicing more of their transactions into pesos ($6.367 / 5.315 \approx 1.20$).

Further zooming in to the origin-product level, we find that the bulk of the within-firm decline in US dollar invoicing arises within continuing origin-product pairs: importers keep

sourcing the same combinations but invoice a smaller share in US dollars over time. By contrast, the rise in Chilean peso use is driven both by the intensive margin-greater invoicing in pesos within continuing origin-product pairs ($3.607/6.367 \approx 0.57$)-and by the extensive margin-the adoption of peso invoicing in new origin-product pairs ($2.131/6.367 \approx 0.33$).

The bottom panel shows that both net currency switching and within-currency adjustments contribute roughly equally to the decline in the within origin-product component of US dollar invoicing. That is, firms both drop the US dollar in favor of alternative currencies (Currency Net Entry) and, conditional on continuing to use the US dollar, reduce the share invoiced in it (Within-Currency). A symmetric interpretation applies for the Chilean peso. Importers increasingly invoice existing origin-product imports in Chilean pesos (Within-Currency), and are more likely to use the domestic currency when adopting a new currency (Currency Net Entry).

Lastly, Table 10 in Appendix A applies the benchmark decomposition to exports. The absence of significant aggregate movements reflects the fact that each of the underlying margins-net firm entry, within-firm adjustments, and between-firm reallocations-shows only modest changes. Thus, aggregate stability does not mask the presence of large but offsetting forces; rather, it reflects the fact that all underlying components changed only minimally, resulting in a muted overall changes in invoicing patterns.

Overall, Table 1 delivers a clear message for interpreting the micro-level drivers of Figure 1. The aggregate decline in the U.S. dollar invoicing share is mainly driven by incumbent importers, which reduce the share of their imports invoiced in U.S. dollars at the origin-product level without significantly changing the quantities they import at that level. A similar pattern emerges for invoicing in Chilean peso: incumbent importers invoice a substantially larger share of their imports in local currency, again without significant changes in imported quantities. Table 1 does not, however, allow us to conclude that there is substitution between the U.S. dollar and the Chilean peso. We therefore turn to the next empirical exercise to address this issue directly.

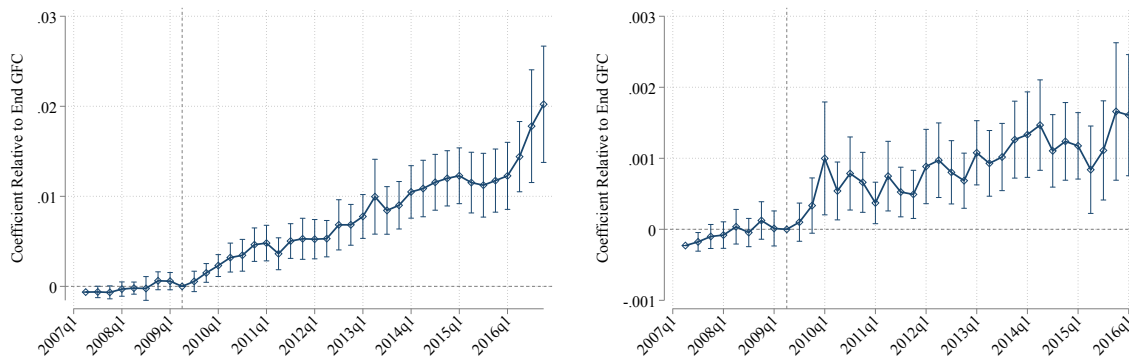
Probability of Invoicing and Switching after the GFC. We estimate whether the likelihood of invoicing in Chilean peso relative to the U.S. dollar has evolved following the GFC. To do so, we estimate the following specification:

$$y_{imt} = \sum_{s=2007Q1}^{2016Q4} \beta_s \cdot \mathbf{1}[t = s] + \gamma_{im} + \epsilon_{imt}, \quad (4)$$

where the dependent variable is a dummy equal to one when a transaction at the imt level is invoiced in Chilean peso and zero when invoiced in U.S. dollars, i.e. $\mathbf{1}\{\text{Peso}_{imt}\}$. As an alternative outcome, we consider a dummy equal to one if a firm invoices a transaction at the im level in U.S. dollars at time $t-1$ and subsequently invoices the same transaction in Chilean peso at time t . The explanatory variables consist of a full set of quarter-year fixed effects spanning the period from 2007Q1 to 2016Q4. Our preferred specification includes $i \times m$

fixed effects, allowing us to exploit within firm–market (HS8-Origin) variation to identify changes in the probability of invoicing in Chilean peso and in the likelihood of switching currency. This level of disaggregation is appropriate for two main reasons. First, it is the level at which we observe meaningful variation in the decomposition exercise. Second, it encompasses the vast majority of transactions and trade flows in our sample. Consistent with this choice, Table 6 in Appendix A shows that multi-currency invoicing within a firm–market–year (*imt*) is rare. Specifically, 96% of *imt* transactions are invoiced in a single currency, and a similar pattern holds when considering the share of import value at the *imt* level. Invoicing in two currencies accounts for the remaining share of transactions and import value.⁸ The prevalence of single-currency invoicing at the *imt* level indicates that firms do not rely on combinations of currencies to implement their sourcing strategies.

Figure 2: Effect of GFC on Probability of Invoicing in Chilean peso



Note: The Figure plots the estimated coefficients from Equation (4). Each point represents the effect of a given quarter on the probability of invoicing in Chilean peso, relative to the end of the GFC (2009Q2). The left panel reports results for the full sample, while the right panel restricts the sample to single-currency importers that switch from US dollar invoicing to Chilean peso invoicing. The vertical dashed line corresponds to the NBER recession period for the end of the GFC. 99 percent confidence interval with standard errors clustered at the importerorigin level are reported. Data are from the Chilean Customs Agency.

We find that both the probability of invoicing in and switching to Chilean peso increased permanently after the GFC. The probability of invoicing in Chilean peso relative to the US dollar increases steadily, reaching 2 percentage points above the pre-GFC level by the first quarter of 2016 (left panel of Figure 2). A similar upward trajectory is observed for the subset of switching firms, where the probability of switching from the US dollar to the Chilean peso is just below 0.2 percentage points higher at the beginning of 2016 than before the GFC (right panel of Figure 2). The effect is quantitatively large considering that, before the GFC, the unconditional probability of invoicing in Chilean peso is 0.7 percent, and the unconditional probability of switching is 0.04 percent.⁹ Importantly, both panels show the absence of pre-trends in firms' invoicing decisions before the GFC, indicating that the structural change in invoicing choices took place around the GFC.

⁸Table 7 in Appendix A shows that similar quantitative patterns hold on the export side.

⁹In line with these findings, Figure 9 in Appendix A shows that the average transition probability from US dollar invoicing to Chilean peso has increase by more than 300 percent after the GFC, while it decreased relative to all other currencies.

Moreover, the switching to local currency invoicing is permanent. Figure 10 in Appendix A shows that, one year after the initial switch from US dollar invoicing to local currency invoicing, roughly 80% of *im*-level transactions continue to be invoiced in Chilean peso. This indicates that firms' invoicing choices are highly persistent over time and the structural change triggered by the GFC produced long-lasting effects.

Robustness: Invoicing Shares over Time Figures 7 and 8 in Appendix A plot the time-series contributions between 2007 and 2019 for each layer of the decomposition for imports invoiced in Chilean peso and US dollar, respectively. The figures show that the decline (increase) in the share of imports invoiced in US dollar (Chilean peso) is not concentrated in a particular year but instead evolves smoothly over time following the GFC. The conclusions drawn in the main section are robust, as they apply not only to the cumulative decomposition reported above, but also to the time-series contributions of each layer at every point in time.

Robustness: Heterogeneity across Products and Origins Figure 11 in Appendix A shows that Chilean peso invoicing dynamics were homogeneous across BEC categories (consumption goods, capital goods, and intermediate goods) and major trading partners (United States, Europe, China, Central-South America, Asia-Middle East, Africa, and Others).

We estimate Equation (4) separately for the top twelve origins and HS2 codes in the sample ranked by number of observations at the *imt* level in our sample. Figures 12 and 13 in Appendix A report the results. The average change in invoicing switching probability is not driven by a specific origin or product imported by Chilean firms, but a common feature across main trading partners and most relevant traded goods.

Moreover, Table 9 in Appendix A presents results from a dynamic Olley-Pakes decomposition with entry and exit at the origin level, abstracting from the product dimension. These results are qualitatively consistent with our benchmark decomposition.

3 A Model of Currency Choice

In this section, we present the theoretical framework used to interpret the empirical invoicing patterns observed among Chilean importers. Our model builds on two baseline contributions in the literature. First, we follow Bahaj and Reis (2020) in modeling within-firm complementarities between the currency used to invoice international trade transactions and the currency in which firms denominate their borrowing costs. Second, drawing on Crowley et al. (2020), we adopt a reduced-form specification of currency-specific fixed costs of invoicing. We extend this fixed-cost structure by introducing sunk costs associated with adopting a new invoicing currency. The presence of sunk costs generates hysteresis in invoicing decisions following temporary shocks to firms financing costs, allowing the model to account for the persistence in invoicing patterns documented in Section 2.

3.1 Environment

A small open economy is populated by a continuum of importing firms indexed by $i \in [0, 1]$. Each firm produces a differentiated good sold domestically and sets prices in domestic currency. Demand for firm i 's output in the domestic market is given by $y_i^d = \left(\frac{p_i^d}{q^d}\right)^{-\theta}$, where p_i^d is the firm's price, q^d is a market-specific demand shifter, and $\theta > 1$ is the constant elasticity of demand.

Production combines domestic labor, l_i , and imported intermediate inputs, x_i , according to a constant-returns-to-scale Cobb–Douglas technology:

$$y_i = A_i l_i^{1-\alpha_i} x_i^{\alpha_i}, \quad (5)$$

where $\alpha_i \in (0, 1)$ denotes the firm-specific share of imported inputs in production, and $A_i \equiv \left(\frac{1}{\alpha_i}\right)^{\alpha_i} \left(\frac{1}{1-\alpha_i}\right)^{1-\alpha_i}$ is a normalization constant. Imported inputs are assumed to be working capital and must be paid for prior to production, whereas labor costs are settled after revenues are realized.

Time is divided into two subperiods within each period: a morning and an evening. In the morning, firms choose prices, which are nominally rigid and remain fixed throughout the period, and finance the imported inputs required for production. Financing can be obtained either in domestic currency d or in a vehicle currency v . In the evening, firms purchase inputs, produce to meet demand at the preset price, collect revenues, and repay their loans.

A key feature of the environment is that the bilateral exchange rate between the domestic currency and the vehicle currency, denoted by s_v , is realized only after financing decisions are made. Consequently, fluctuations in the exchange rate affect the realized cost of imported inputs and can generate deviations between preset prices and ex post marginal costs.

Firms face the following production technology for imported inputs x_i :

$$x_i = \min_{\eta_i} \left(\frac{x_i^d}{\eta_i}, \frac{x_i^v}{1 - \eta_i} \right). \quad (6)$$

By choosing η_i , the firm determines the relative shares of two imported inputs: x_i^d , denominated in domestic currency d , and x_i^v , denominated in the vehicle currency v . It follows from standard theory that firms optimally choose corner solutions for invoicing. In particular, imported inputs are entirely invoiced in either domestic or vehicle currency, so that $\eta_i^* \in \{0, 1\}$. In the corner solutions, if $\eta_i^* = 0$, the firm invoices imported inputs exclusively in the vehicle currency, whereas if $\eta_i^* = 1$, it invoices imported inputs entirely in domestic currency.

The firm must borrow to finance imported inputs and can choose between domestic and foreign currency borrowing. In principle, a firm could finance its imported inputs in a currency different from the one used to invoice international trade transactions. Following [Bahaj and Reis \(2020\)](#), however, we restrict attention to the case in which the financing currency coincides with the invoicing currency of imported inputs. This simplification is without loss of generality, as it arises optimally: firms prefer to align financing and invoicing currencies

to minimize the exchange rate risk created by currency mismatches. Hence, in the model the currency chosen to invoice borrowing costs is equivalent to the currency used to invoice imported inputs.

The firm faces two financing options: (i) borrowing one unit of domestic currency and repay $(1 + i_d)$, or (ii) borrowing one unit of vehicle currency and repay $(1 + i_v)$. While interest rates are known at the time of borrowing, the exchange rate between the domestic and vehicle currencies, s_v , remains uncertain.

Given the cost structure described above, the *ex post* marginal cost of production for firm i , expressed in domestic currency, is:

$$C_i(s^v, \eta_i) = \frac{1}{A_i} \left[\frac{\eta_i \times \rho_d(1 + i_d) + (1 - \eta_i) \times \rho_v s_v(1 + i_v)}{\alpha_i} \right]^{\alpha_i} \left[\frac{w}{1 - \alpha_i} \right]^{1 - \alpha_i}, \quad (7)$$

where w denotes the domestic wage rate, and ρ_d and ρ_v are the prices of domestic and imported intermediate inputs expressed in domestic and vehicle currency, respectively. All prices and borrowing costs are taken as exogenous and common across firms.

Lastly, we turn to the final key cost component: currency- and firm-specific fixed costs. Firms invoicing international transactions in currency j incur a fixed cost F_i^j . Following [Crowley et al. \(2020\)](#), we assume that this cost decreases with the number of firms using currency j . We extend their specification by introducing sunk costs in the adoption of new currencies, implying that a firm's fixed cost of using currency j depends on its invoicing choice in the previous period.

Formally, the reduced-form representation of firm i 's fixed cost of invoicing in currency j at time t is:

$$F_{it}^j = f_{it}^j - \gamma^j \omega_{t-1}^j(N_{-it}), \quad (8)$$

where $\omega_{t-1}^j(N_{-it})$ denotes the share of importers using currency j in period $t - 1$. The firm-specific component is given by

$$f_{it}^j = \begin{cases} \kappa_0^j & \text{if firm } i \text{ uses currency } j \text{ for the first time,} \\ \kappa_1^j & \text{if firm } i \text{ has previously used currency } j, \end{cases}$$

with $\kappa_1^j < \kappa_0^j$. The difference $\kappa_0^j - \kappa_1^j$ represents the sunk entry cost of invoicing in a new currency, capturing one-time managerial costs such as setting up foreign-currency bank accounts.¹⁰

The second term in Equation (8) captures externalities in adoption: the fixed cost decreases as more firms choose currency j , reflecting strategic complementarities in invoicing ([Amiti et al., 2019](#); [Crowley et al., 2020](#)).¹¹ The strength of these complementarities is gov-

¹⁰We assume that the firm-specific component depends only on the previous periods currency choice. Richer specifications could allow a longer invoicing history to matter, leading to gradual cost accumulation. See [Alessandria et al. \(2021\)](#) for a review of sunk costs in the trade literature.

¹¹In our framework, complementarities operate across firms. Empirical evidence in Section 2 shows that invoicing switching is not driven by specific destination-product pairs within firms. Nonetheless, the model can

erned by γ^j , which satisfies the stability condition $0 < \gamma^j < f_{it}^j < 1$. The core idea of complementarities is that usage rises as overall utilization increases. One practical rationale is that firms may prefer to invoice in the same currency as their competitors to maintain similar exchange rate pass-through to import prices.

3.2 Working Capital Invoicing Choice

In this section, we solve for both the optimal invoicing decision for imported inputs and the optimal domestic pricing decision. Crucially, we obtain key predictions that we will later test directly in the data.

Optimal Invoicing Decision Firms are assumed to be risk-neutral and aim to maximize expected profits. They form expectations about the future realization of the exchange rate between the domestic currency and the vehicle currency, s_v . Fluctuations in exchange rate realizations affect the ex-post cost of borrowing in the vehicle currency, generating deviations from the optimal markup over marginal costs and reducing profits.

Given the isoelastic demand function and the imported-input technology in Equation (6), firm i maximizes ex-ante profits:¹²

$$\max_{\eta_i, p_i^d} \mathbb{E} [\pi_i^j] = p_i^d (p_i^d)^{-\theta} - \mathbb{E} [C_i(s_v, \eta_i)] (p_i^d)^{-\theta} - \eta_i F_i^d - (1 - \eta_i) F_i^v. \quad (9)$$

For a given η_i , the optimal domestic price is $p_i^{d*} = \frac{\theta}{\theta-1} \mathbb{E} [C_i(s_v, \eta_i)]$. Given p_i^{d*} , firm i chooses to invoice imported inputs in the vehicle currency ($\eta_i^* = 0$) if the following condition holds:

$$\eta_i^* = 0 \quad \text{iff} \quad \mathbb{E} [(\rho_v(1 + i_v)s_v)^{\alpha_i}]^{1-\theta} - [\rho_d(1 + i_d)]^{\alpha_i(1-\theta)} > \tau_i (F_i^v - F_i^d), \quad (10)$$

where $\tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}$. See Appendix B for the derivation of Equation (10).

From inspecting Equation (10) and assuming that the exchange rate s_v is log-normally distributed with mean μ and variance Σ , Appendix B proves the following result:¹³

Proposition 1. Higher borrowing costs in vehicle currency, i_v , lowers the probability of invoicing in vehicle currency, *ceteris paribus*.

Invoicing imported inputs in vehicle currency v is optimal when its financing costs are sufficiently low to compensate for exchange rate uncertainty, after accounting for relative input costs and fixed costs of invoicing. Firms maintain a constant markup over marginal costs, so vehicle-currency invoicing implies that ex-post marginal costs depend on the realized exchange rate, potentially generating costly deviations from the optimal markup. Consequently,

be extended to multi-market firms, capturing both within- and across-firm externalities.

¹²Without loss of generality, the demand shifter q_d is normalized to one.

¹³While log-normality allows for closed-form solutions, the same intuition holds using a second-order approximation with a general distribution.

firms choose vehicle-currency invoicing only when its financing cost advantage outweighs the risk of markup misalignment.

Appendix B proves that the trade-off between cheaper financing option and costly departure from the optimal markup depends on the exposure of the marginal cost to exchange rates fluctuations, α_i , capturing firm's operational hedging motive (Amiti et al., 2022):

Proposition 2. For sufficiently high exchange rate volatility, an increase in the cost of financing in vehicle currency, i_v , reduces the probability that a firm invoices in vehicle currency. This effect is stronger for firms with a higher share of imported inputs, α_i .

This proposition reflects the natural hedging behavior of firms in our setup. Since all revenues are assumed to be in domestic currency, firms optimally prefer to match the currency of their imported-input costs with that of their revenues. The incentive to do so is stronger the larger the share of imported inputs, α_i , because these firms are more exposed to fluctuations in vehicle-currency financing costs. Consequently, when i_v rises, the probability of invoicing in vehicle currency declines more sharply for firms with higher α_i , minimizing costly deviations from the profit-maximizing price. Proposition 2 formalizes this intuition.

Appendix B proves the following on the role of strategic complementarities:

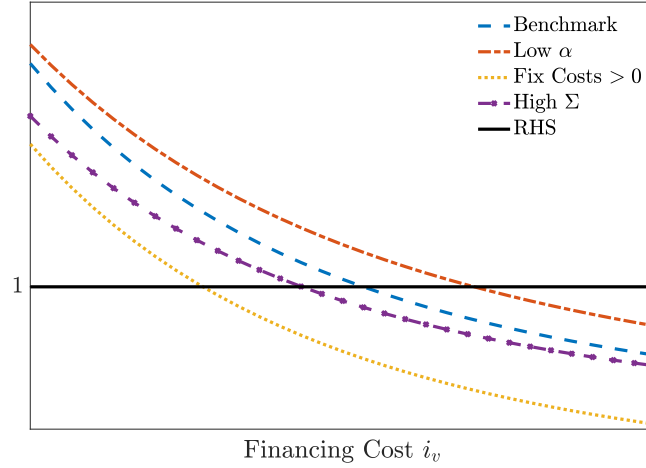
Proposition 3. Stronger strategic complementarities that is, lower fixed costs of invoicing reduce the probability that a firm switches to domestic-currency invoicing when the cost of financing in vehicle currency, i_v , rises.

The term on the right-hand side of Equation (10) captures how fixed costs of currency use affect the invoicing choice.¹⁴ The firm optimally invoices in vehicle currency only when its financing cost advantage is large enough to offset both the deviation from the optimal constant markup and the differential in fixed costs of invoicing. Thus, lower fixed costs of invoicing in vehicle currency, F_v , influenced by both the intensity of strategic complementarities in invoicing and by sunk costs, make vehicle currency borrowing more convenient even for higher levels of i_v .

Figure 3 graphically represents the optimal decision rule in Equation (10) and illustrates how it depends on the firm-specific share of intermediate inputs in production α_i , exchange rate uncertainty Σ , and the differential in currency-specific fixed costs ($F_i^v - F_i^d$). Expected profits under domestic currency financing are normalized to one and independent of i_v (horizontal line). By contrast, expected profits when invoicing working capital in vehicle currency decline with i_v , since higher vehicle interest rates raise marginal costs (downward-sloping line). The intersection of the two curves determines the cut-off value of i_v above which it is no longer optimal to invoice in vehicle currency, that is, $\eta^* = 1$.

¹⁴In deriving the optimal invoicing choice, we assume that firms do not internalize the effect of their choice on the aggregate invoicing share and, consequently, on fixed costs (Equation (8)).

Figure 3: Invoicing Choice - Comparative Statics



Note: Figure 3 plots the optimality condition in Equation (10), $\mathbb{E}[(\rho_v(1+i_v)s_v)^{\alpha_i}]^{1-\theta} - \tau_i(F_i^v - F_i^d) > [\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, for different levels of the vehicle currency interest rate i_v , assuming $s_v \sim \text{Log-Normal}(\mu, \Sigma)$. The RHS, $[\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, is normalized to one. The blue line shows the benchmark calibration ($w = 0.2$, $\rho_v = \rho_d = 1$, $\theta = 5$, $\mu = 0$, $\Sigma = 1$, $\alpha_i = 0.8$, and $F_i^v - F_i^d = 0$). The other lines illustrate comparative statics relative to the benchmark: lower input share $\alpha_i = 0.7$ (red), higher fixed-cost differential $F_i^v - F_i^d = 1$ (yellow), and higher exchange rate variance $\Sigma = 1.05$ (purple).

Figure 3 illustrates the comparative statics of the optimal invoicing decision. Consistent with Proposition 2, a lower share of imported inputs, α_i , shifts the expected-profit curve for vehicle-currency invoicing to the right, increasing the likelihood that firms choose vehicle currency. In line with Proposition 3, a larger fixed-cost differential, $F_i^v - F_i^d$, shifts the curve leftward, making vehicle-currency invoicing less attractive. Finally, higher exchange rate volatility, Σ , raises the cost of deviating from the optimal markup, shifting the vehicle-currency expected-profit curve leftward and further reducing the probability that firms invoice in vehicle currency.

Lastly, to further analyze the mechanism, we can further assume that fixed costs are equal across currencies, $F_i^v = F_i^d$ for all i . Under this additional assumption, the cutoff condition can be rewritten as:

$$\eta_i^* = 0 \quad \text{iff} \quad \underbrace{i_v - i_d + [\mu + 0.5 \alpha_i \Sigma]}_{\text{UIP Deviation}} < \log \left(\frac{\rho_d}{\rho_v} \right), \quad (11)$$

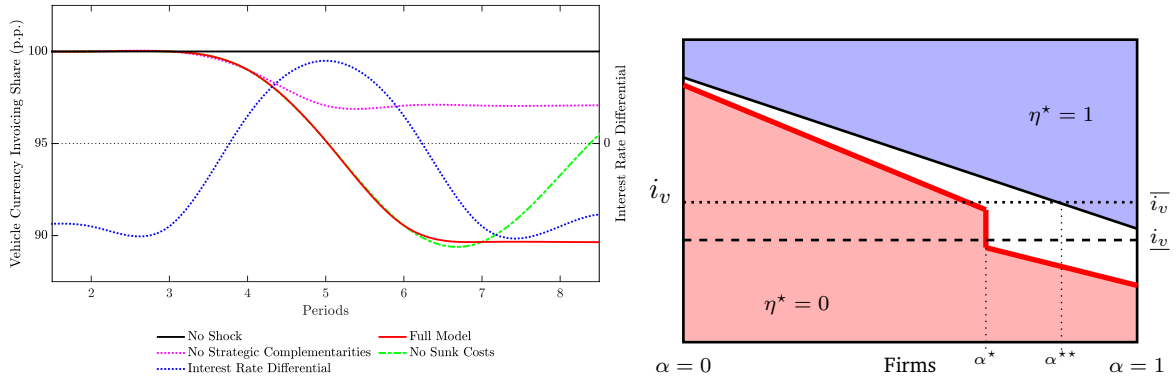
where μ and Σ are the mean and variance of $s_v \sim \text{Log-Normal}(\mu, \Sigma)$. Equation (11) offers additional insights from the model: firm-specific UIP deviations shape importers currency choices. The left hand side of Equation (11) shows that we can recast the key predictions of the models in terms of UIP fluctuations, drawing a direct link between UIP deviations and firm-level invoicing decisions and, in the aggregate, the dominance of a currency.

3.3 Temporary Shocks, Invoicing Dynamics and Hysteresis

Next, we qualitatively assess the effects of a temporary rise in the cost of finance in vehicle currency i_v – or, equivalently, an increase in UIP deviations – that permanently reverts to its original level after a few periods. We show that our theory can rationalize the main empirical findings of Section 2: the long lasting effects on the individual and aggregate import invoicing patterns. Moreover, firms that are larger and more exposed to exchange rate are the first to switch to domestic currency invoicing. Figure 4 displays these dynamics.

The firm-level invoicing patterns presented in Section 2 support the assumption that all firms initially invoice in vehicle currency and have no prior experience invoicing in domestic currency. As a result, firms must pay a sunk cost to switch to domestic currency invoicing. Moreover, we assume that the share of imported goods used in production, α_i , is uniformly distributed: $\alpha_i \sim U[0, 1]$.¹⁵

Figure 4: Shock to Vehicle Currency Financing



Note: The right panel plots the optimal invoicing decision for each level of i_v (y-axis) conditional on the level of α (x-axis). The left panel plots the dynamics of the aggregate shares of US dollar invoicing. The red line represents the full model; the green one abstracts away from sunk costs; the purple one abstracts away from strategic complementarities. The blue line represents the dynamics of the relative costs of financing $i_v - i_d$. We use the following symbolic calibration: we normalize ρ_v and ρ_d to one; the elasticity of demand θ is set equal to 5; domestic wage is set to 0.2 (Saravia and Voigtländer, 2012); α_i are uniformly distributed; without loss of generality, we consider the following specification for the fixed costs of invoicing in US dollar: $F^v - F^d = k_0^v - f^d - \gamma^v \omega_{t-1}^v - \gamma_1^v (\omega_{t-2}^v - \omega_{t-1}^v)$, where γ^v is set equal to 0.7, $\gamma^d = 0$, $\gamma_1 = 2.5$, $k_0^v = 1$, and $f^d = \{k_1^d, k_0^d\} = \{0.2, 0\}$; we consider a 3% increase in $i_v - i_d$.

Short-run response The right panel of Figure 4 graphically represents each firm's invoicing choice. On the horizontal axis, firms are ranked in an ascending order based on their share of imported inputs, α_i . The vertical axis represents the cost of financing in vehicle currency, i_v . For any given level of i_v , there is a threshold $\tilde{\alpha}$ such that firms with $\alpha_i < \tilde{\alpha}$ invoice in domestic currency ($\eta^* = 1$), while firms with $\alpha_i > \tilde{\alpha}$ invoice in vehicle currency ($\eta^* = 0$). The downward-sloped line dividing the box in two areas – domestic (blue area) and vehicle (non-blue area) currency financing – represents the locus of $\tilde{\alpha}$ as a function of i_v .¹⁶

Following the rise in the cost of financing in vehicle currency, larger and more exposed

¹⁵The main insights are qualitatively unchanged regardless of the distributional assumption on α_i .

¹⁶For instance, in the special case where $s_v \sim \text{Log-Normal}(\mu, \Sigma)$, the cutoff $\tilde{\alpha}$ is determined by Equation (11) once we solve explicitly for α_i .

firms are the first to switch to domestic invoicing. The initial cost of financing in vehicle currency, i_v , is low enough that it is optimal for all firms to invoice in vehicle currency. This is represented by i_v (dash horizontal line) lying below the initial locus of $\tilde{\alpha}$ (solid black line). When i_v increases from the initial level i_v to a higher level $\bar{i}_v$ – meaning foreign invoicing becomes less convenient – a mass of firms with $\alpha_i > \alpha^{**}$ begin borrowing and invoicing in local currency. Notably, the first firms to switch are those with high α_i , which represents a sufficient statistics for both their unhedged exposure to vehicle currency and their market share in the imported inputs market.¹⁷

The presence of strategic complementarities creates a positive feedback in domestic currency invoicing, further expanding its use in imports. The left panel of Figure 4 shows that the aggregate share of imports invoiced in vehicle currency continues to decline even after the cost of financing in vehicle currency, i_v , starts to revert. The use of domestic currency by a group of firms after the shock reduces the fixed cost of invoicing in domestic currency for others, increasing the likelihood that more firms will adopt it. Firms with lower α_i value, relative to the initial switchers, begin to use domestic currency to invoice their imported inputs as domestic invoicing becomes more convenient for them.¹⁸

Hysteresis in invoicing The presence of sunk cost in invoicing implies that once a firm switches to domestic currency invoicing, the relative convenience of domestic versus vehicle currency invoicing is permanently altered. As a result, the sunk-cost model generates invoicing hysteresis: even if i_v permanently reverts to the original level i_v , firms do not switch back to vehicle currency invoicing. Specifically, the difference between the fixed costs of invoicing in vehicle and domestic currency, $F^v - F^d$ in Equation (10), decreases, making domestic currency invoicing permanently more attractive than vehicle currency invoicing.

Graphically, this is represented by a downward shift in the threshold between domestic and vehicle currency invoicing for firms that begin invoicing in domestic currency (i.e. those with $\alpha_i > \alpha^*$). The area where domestic invoicing is optimal ($\eta^* = 1$) extends to include the area above the solid red line (i.e. both the white area and the original blue area). Thus, the new cut-off rule implies that for firms with $\alpha_i > \alpha^*$, it remains optimal to invoice in domestic currency even when ϵ reverts to the original level, i_v .¹⁹

At the aggregate level, this implies that the share of imports invoiced in domestic currency does not return to one after i_v reverts to the original level, which is consistent with

¹⁷The former follows immediately from the fact that firms are importers only, with revenues denominated in domestic currency. The latter follows from the fact that firms are assumed to have identical size. Thus, higher α_i implies more imported inputs and, thus, a larger market share in the input market.

¹⁸In the right panel of Figure 4, firms with $\alpha^* < \alpha_i < \alpha^{**}$ gradually switch to domestic currency invoicing as fixed costs decrease. At the aggregate level, the pink dashed line in the left panel shows that, when the i_v starts to revert, the share of vehicle currency invoicing stops decreasing in the absence of strategic complementarities.

¹⁹Notice that domestic currency invoicing becomes more appealing even for firms that have not yet paid the sunk cost, i.e. firms with $\alpha_i < \alpha^*$. This is because the fixed cost of invoicing in domestic currency decreases also for them, due to strategic complementarities. Relative to the initial steady state, where all invoicing was in vehicle currency, the fixed cost of invoicing in vehicle currency is now higher since a subset of firms has permanently switched to domestic invoicing. The higher fixed cost of invoicing in vehicle currency F^v makes domestic invoicing more competitive for all firms.

the observed data. The absence of any partial reversion in the aggregate share of imports invoiced in domestic currency suggests that the sunk cost associated with domestic currency invoicing is significant enough to make the switch permanent. The green dashed line in the left panel shows that, if the sunk cost were smaller, the aggregate share of invoicing could (partially) increase back, as it might become optimal for some firms to revert to vehicle currency invoicing. This would occur if the decrease in the fixed costs of domestic invoicing, F^d , is insufficient to make domestic currency invoicing optimal at the original level of $i_v, \underline{i}_v$.

4 Empirical Evidence

In this section, we return to the data to test the model's predictions. We begin by documenting that the US dollar-peso UIP dynamic is consistent with the mechanism that incentivizes firms to switch from vehicle-currency invoicing to domestic-currency invoicing (Proposition 1). Next, we show that, as the cost of borrowing in U.S. dollars rises relative to domestic currency, the probability of invoicing in Chilean pesos decreases with the share of imports invoiced in U.S. dollars (Proposition 2) and decreases with the strength of strategic complementarities in U.S. dollar invoicing (Proposition 3). These coefficients are identified using cross-sectional variation across Chilean importers.

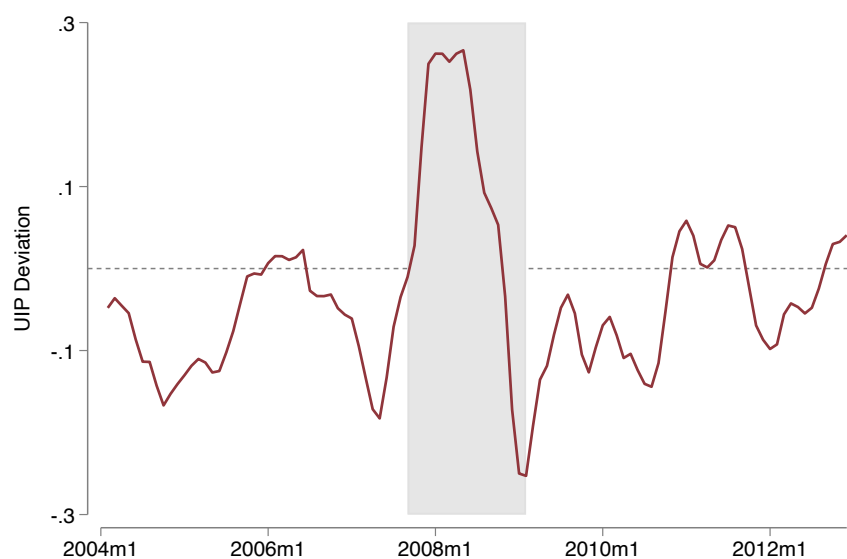
US Dollar-Chilean Peso UIP dynamics. Following the theoretical framework, we construct realized peso-dollar UIP deviations as a time-varying proxy for the financing-cost differential between local and vehicle currencies, which we use to validate the models predictions.²⁰ Specifically, we define the realized log UIP deviation between the Chilean peso and the US dollar as

$$\text{UIP}_{t+1}^{\$, \text{CLP}} = \underbrace{i_t^{\$} - i_t^{\text{CLP}}}_{\Delta \text{Interest Rates}} + \underbrace{s_{t+1} - s_t}_{\Delta \text{Exchange Rate}}, \quad (12)$$

where $i_t^{\$} \approx \log(1 + i_t^{\$})$ and $i_t^{\text{CLP}} \approx \log(1 + i_t^{\text{CLP}})$ denote commercial lending rates on short-term (less than one year) loans denominated in US dollars and Chilean pesos, respectively. We use data on short-term bank lending rates in both currencies from the Chilean Financial Authority responsible for supervising the banking system. These rates capture the interest cost faced by a domestic firm borrowing from domestic banks, either in Chilean pesos or in US dollars. The data cover the period around the GFC and are available until 2014. Here, s_t denotes the log bilateral spot exchange rate (CLP per USD), and s_{t+1} is the realized exchange rate one year ahead. A negative UIP deviation ($\text{UIP}_{t+1} < 0$) indicates that borrowing in local currency is more costly than borrowing in US dollars, while a positive deviation implies the opposite.

²⁰Technically, the model implies that the UIP premium is the closest proxy for the differential in borrowing costs. Unfortunately, data limitations restrict the yearly sample used to identify the coefficients in Equation (13) to start in 2007. In Figure 15 in Appendix A.6, we show that when the UIP deviation in Equation (12) is constructed using the one-year-ahead forecast exchange rate, the two measures display qualitatively similar dynamics.

Figure 5: Realized UIP Between the US Dollar and the Chilean Peso



Note: The figure plots the one-year realized UIP deviation between the Chilean peso and the US dollar, constructed using Equation (12). A positive value indicates that borrowing in Chilean pesos is relatively cheaper than borrowing in US dollars. Interest rates are measured using commercial lending rates on short-term (less than one year) loans denominated in Chilean pesos and US dollars offered by domestic banks. Data are monthly and smoothed using a two-month rolling window.

During the GFC, borrowing in Chilean pesos became temporarily cheaper than borrowing in US dollars, reversing the pattern observed in normal times. Figure 5 shows that outside the crisis period, realized UIP deviations are negative and fluctuate around 10 percentage points, indicating that borrowing in local currency is typically more expensive than borrowing in US dollars.²¹ In contrast, during the GFC the average realized UIP deviation increased sharply, turned positive, and peaked at nearly 30 percentage points, implying a sudden rise in the cost of dollar-denominated borrowing. This pattern is consistent with a shortage of dollar funding at the height of the financial crisis (Ahn et al., 2011; Amiti and Weinstein, 2011). By 2009, the relative cost of financing had reverted to its negative, pre-crisis level.²²

The transitory nature of the shock to the relative cost of financing in local and vehicle currencies is consistent with the theoretical mechanism required to generate a persistent decline in the share of vehicle-currency invoicing. Importantly, the timing and magnitude of these movements do not coincide with changes in banking or capital-market regulation. Major reforms to liberalize the financial sector, enhance supervision, and deepen international integration—such as improvements in access to foreign-exchange hedging instruments—were implemented only after the crisis period (Fund, 2010). Similarly, Chile adopted key prudential regulations later than other advanced and OECD economies.²³

²¹The magnitude is consistent with existing evidence in the literature; see Kalemli-Ozcan and Varela (2021).

²²These dynamics are in line with previous evidence on currency-specific borrowing costs in Chile; see Vial et al. (2020) and Betancour et al. (2006).

²³See Madeira and Olivares (2021) and <https://www.sipa.columbia.edu/sites/default/files/2023-02/Macroprudential-Policies-A-View-From-Chile.pdf>. Basel III was implemented in 2019; see <https://www.cmfchile.cl/portal/principal/613/w3-article-50324.html>.

4.1 Firm-Level Evidence

Using cross-sectional variation, we test the key predictions of the theoretical framework.

Cross-Sectional Analysis. We start by estimating the following specification:

$$\begin{aligned} Peso_{fpot} = & \beta_0 + \beta_1 UIP_t^{\$,CLP} + \beta_2 NH_{ft}^{\$} + \beta_3 SC_{(-f)ost}^{\$} + \\ & + \beta_4 NH_{ft}^{\$} \times UIP_t^{\$,CLP} + \beta_5 SC_{(-f)ost}^{\$} \times UIP_t^{\$,CLP} + \\ & + FE_{fpo} + \nu_{fpot}, \end{aligned} \quad (13)$$

where the dependent variable is a dummy variable that takes value equal one when the transaction at the firm-origin-product(HS8)-time (f_{opt}) level is invoiced in Chilean peso and zero if in US dollar. $UIP_t^{\$,CLP}$ is the constructed following Equation (12) and represented in Figure 5, proxying the relative cost of financing in local and vehicle currency. In the model, α_i measures the share of imported inputs in production that are invoiced in a given currency. We can construct an analogous measure in the data. However, we cannot use this variable directly when estimating Equation (13) because of a clear reverse causality concern. We therefore follow the previous literature (Amiti et al., 2022; Crowley et al., 2024) and use the share of firm f exports invoiced in US dollar at time t as an alternative explanatory variable denoted as $NH_{ft}^{\$}$. Lastly, the term $SC_{(-f)ost}^{\$}$ captures strategic complementarities in invoicing. We measure strategic complementarities in invoicing as in Crowley et al. (2020):

$$SC_{(-f)ost}^{\$} = \frac{\sum_{k \neq f} \text{ToT Imp}_{kost}^{\$}}{\sum_c \sum_{k \neq f} \text{ToT Imp}_{kost}^c} \quad (14)$$

where $(-f)$ is the set of competitors of firm f in sector s (HS4) who import from origin o at time t . Equation (14) measures the share of total imports of competitors at the industry (HS4) and origin level invoiced in US dollar. Clearly, the invoicing choice of firm f is endogenous to the choice of its competitors $(-f)$. Following previous work (Amiti et al., 2022; Crowley et al., 2020), we construct the following instrumental variables to correct for the existence of possible endogeneity:

$$\widehat{SC}_{(-f)ost}^{\$} = \sum_{k \neq f} \frac{S_{kost}}{1 - S_{kost}} \times \frac{\text{ToT Exp}_{kt}^{\$}}{\sum_c \text{ToT Exp}_{kt}^c}, \quad (15)$$

where $S_{kost} = \frac{\text{ToT Imp}_{kost}}{\sum_f \text{ToT Imp}_{fost}}$ is therefore the import market share of firm k from origin o , sector s (HS4), and time t level among all Chilean importers, and the last ratio is the share of firm k 's total exports that are invoiced in US dollars.

The specification in Equation (13) allows us to empirically test the key proposition of the model by exploiting cross-sectional variation. Proposition 1 predicts that an increase in borrowing costs denominated in vehicle currency raises the probability that imports are invoiced in local currency, implying a positive coefficient β_1 . Proposition 2 predicts a negative coefficient

Table 2: Model Validation: Cross-Sectional Analysis

	OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
$UIP_t^{$,PESO}$	0.028*** (0.001)	0.027*** (0.001)	0.026*** (0.001)	-	-	-
$NH_{ft}^{\$}$		-0.018*** (0.001)	-0.017*** (0.001)	-0.015*** (0.001)	-0.015*** (0.001)	
$SC_{kt}^{\$}$			-0.033*** (0.002)	-0.193*** (0.005)	-0.122*** (0.003)	-0.193*** (0.005)
$SC_{kt}^{\$} \times UIP_t^{$,PESO}$				-0.186*** (0.012)	-0.169*** (0.011)	-0.188*** (0.012)
$NH_{ft}^{\$} \times UIP_t^{$,PESO}$				-0.044*** (0.014)	-0.054*** (0.015)	0.243*** (0.082)
Firm-Origin-HS8	Yes	Yes	Yes	Yes	No	Yes
Year	No	No	No	Yes	Yes	Yes
Firm-Origin	No	No	No	No	Yes	No
Firm-HS8	No	No	No	No	Yes	No
Observations	1001124	1001124	965928	965928	1158217	965928

Note: The Table reports the coefficients from specification in Equation (13). The dependent variable is a dummy variable equal to 1 when the f_{opt} transaction is invoiced in Chilean peso and 0 otherwise. IV columns refers to the case in which the strategic complementarity index is instrumented with the import-weighted average US dollar exposure of all competitors in a given sector(HS4)-origin pair as in Equation (15). We augment the specification with time fixed effects in Columns (4) - (6), thus the coefficient β_1 cannot be identified. In Column (6) we proxy the exposure to financing costs using importer size-measured by firm f 's share of aggregate imports. Standard errors are clustered at firm-product-origin level. We use data from 2004 until 2014 from [Garcia-Marin et al. \(2019\)](#) because of data limitation on our UIP measure. Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

cient for β_4 . Indeed, given that US dollar borrowing is relatively more expensive, the stronger is the incentive to match revenues and costs in US dollars, the lower will be the probability that firms invoice imports in domestic currency. Thus, a higher share of exports denominated in US dollars is associated with a lower probability that imports are invoiced in pesos. Proposition 3 further predicts that the effect of changes in vehicle-currency borrowing costs is weaker in the presence of stronger strategic complementarities in invoicing. Accordingly, β_5 is expected to be negative.

Table 2 reports the estimates from Equation (13) and shows that the data support the key predictions of the model. Consistent with Proposition 1, Columns (1) - (3) show that the coefficient on the UIP deviations, β_1 , is positive and significant. A one-percentage point increase in the cost of financing in US dollar-proxied by the UIP deviation-raises the probability of invoicing in Chilean peso by approximately 2.6-2.8 basis points. This result aligns with the models prediction that an increase in the expected cost of borrowing in US dollar relative to the Chilean peso strengthens firms' incentives to invoice in domestic currency. Given the roughly 40-percentage-point increase in our UIP measure during the GFC (Figure 5), the estimated coefficient implies that the probability of invoicing in local currency increased by 1-1.5 percentage points, accounting for roughly half of the aggregate dynamics shown in Figure 2.

The main specification in Column (4) provides evidence in line with Propositions 2 and 3. The interaction terms between the borrowing-cost differential and operational hedging ($NH_{ft}^{\$}$) and strategic complementarities in invoicing ($SC_{(-f)ost}^{\$}$) are both negative and precisely estimated.²⁴ These results imply that, conditional on an increase in US dollar borrowing costs, firms are less likely to switch to local-currency invoicing when their competitors invoice a larger share of imports in US dollars and when they themselves invoice a larger share of exports in US dollars. In line with the theoretical framework, strategic complementarities in vehicle-currency invoicing dampen the incentive to switch, while greater exposure to US dollar invoicing on the export side induces firms to align import invoicing with their export currency, further reducing the likelihood of switching.

Finally, Column (5) shows that the results are quantitatively robust to alternative sources of identifying variation. Specifically, we replace firm-product-origin fixed effects with a combination of firm-origin and firm-product fixed effects, obtaining estimates that are similar to those in the main specification. Column (6) further confirms the key intuition when we proxy exposure to financing costs, α_i , using importer size-measured by firm f 's share of aggregate imports. In this case, Proposition 2 predicts a positive coefficient, which is borne out in the data.

4.2 Empirical Validation of Dynamic Model Predictions

Event Study Design. We provide additional econometric evidence on the heterogeneous effects due to exposure and strategic complementarities using event study specifications. We estimate the following specification:

$$Pes_{opt} = \sum_{s=2007Q1}^{2015Q4} \left[\mathbf{1}[t = s] \times \left(\beta_s \cdot X_{f,2007Q4} \cdot \text{UIP}_{\text{GFC}}^{\$,CLP} \right) \right] + FE_{fpo} + \epsilon_{fpo}, \quad (16)$$

where the dependent variable is still a dummy variable that takes value equal one when the transaction at the firm-origin-product(HS8)-time level is invoiced in Chilean peso and zero if in US dollar; $\mathbf{1}[t = s]$ is an indicator equal to 1 in quarter s and 0 otherwise.

The product $X_{f,2007Q4} \cdot \text{UIP}_{\text{GFC}}^{\$,CLP}$ captures the intensity of the shock suffered by a firm f around the GFC, where we fix the exposure at the pre-shock level and we interact it with the GFC shock. The term $\text{UIP}_{\text{GFC}}^{\$,CLP}$ is the change in UIP deviation between USD and CLP over the recession window (2007Q4-2008Q4). The firm-level exposure $X_{f,2007Q4}$ is fixed at the 2007Q4 level. The term $X_{f,2007Q4}$ is the share of imports denominated in US dollar when assessing the role of firm exposure and operational hedging (Proposition 2). The term $X_{f,2007Q4}$ is instead the sector-origin index of strategic complementarity in US dollar invoicing from Equation (14) when assessing the role of strategic complementarities (Proposition 3). Crucially, $X_{f,2007Q4}$ is fixed before the crisis, so it is not affected by firms subsequent invoicing

²⁴The main specification includes time fixed effects to absorb aggregate dynamics. As a result, we identify only heterogeneous effects, not the average effect of UIP deviations. Table 2 also reports the average effects of the operational-hedging and strategic-complementarity measures, which are consistent with existing literature.

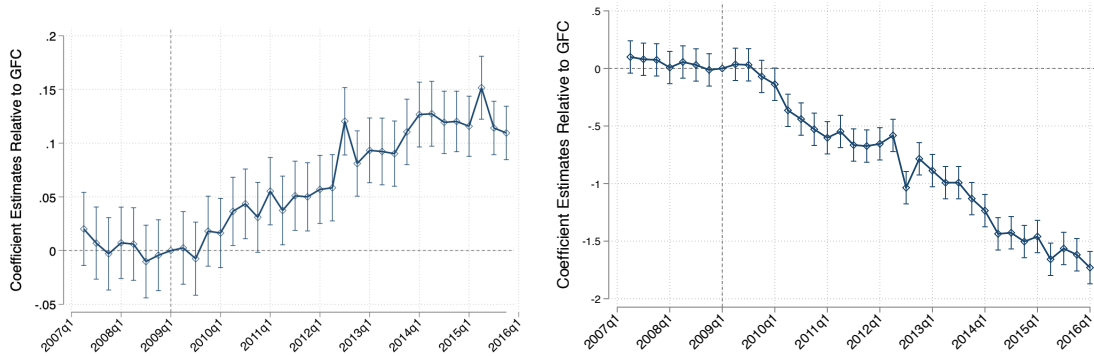
decisions. This pre-determined exposure allows us to avoid the reverse-causality concerns that arise in the cross-sectional regressions (Equation (13)).

The identification strategy for the event study is straightforward. We exploit the unexpected nature of the sharp movements in UIP deviations during the GFC, and the resulting heterogeneity in exposure across firms driven by their pre-crisis characteristics. Identification comes from interacting time indicators with firms pre-GFC exposure to the UIP shock, while saturating the specification with firmproductorigin fixed effects.

Accordingly, the coefficients β_s capture how the impact of the GFC-induced increase in the relative cost of dollar financing on invoicing choices evolves over time as a function of firms pre-shock exposure. In particular, β_s measures the differential change at quarter s in the probability of invoicing in Chilean peso associated with higher exposure to the UIP shock prior to the crisis.

The theory predicts that, following an increase in the cost of borrowing in US dollars, firms with greater pre-GFC exposure to dollar-denominated imports are more likely to switch to domestic currency invoicing. Consistent with Proposition 2, this implies positive values of β_s . In contrast, firms operating in environments characterized by stronger strategic complementarities in US dollar invoicing face higher effective switching costs and are therefore less likely to move toward peso invoicing when dollar financing costs rise. In line with Proposition 3, this mechanism translates into negative values of β_s .

Figure 6: Event Study



Note: The Figure plots the coefficients β_s from the event study specification in Equation (16), capturing the differential increase in the probability of Chilean peso invoicing by more exposed firms. The left panel measures exposure in terms of operational hedging (Proposition 2), proxied by the share of imports denominated in US dollar. In the right panel we use the strategic complementarity measure from Equation (14) (Proposition 3). In the left (right) panel, we absorb firm-product-origin and time (firm-product-origin) fixed effects. Coefficient in 2009Q1 is normalized to zero. Data are from Chilean Customs from 2007 until 2016. We report 90 percent confidence intervals. Standard errors are clustered at firm-product-origin level.

Figure 16 shows that the event-study estimates are consistent with these predictions and consistent with the dynamics in our simulation (Section 3.3). The effects emerge during the crisis period, persist well beyond the reversion of UIP deviations to pre-crisis levels, and display a monotonic pattern over time. Importantly, we find no evidence of pre-trends or anticipation effects, supporting the validity of the empirical design.

In Appendix A.6, we show that the key qualitatively results hold when we use alternative

identifying variation, annual data from 2004 from [Garcia-Marin et al. \(2019\)](#), and discrete treatment separating the top quartile of the exposure from the rest of the distribution.

Sunk Costs in Invoicing We provide evidence, consistent with the international trade literature, that supports the presence of sunk costs in invoicing decisions, which is a key mechanism explaining the hysteresis in currency choice. The key implication of sunk costs is that the decision to invoice in a specific currency is more likely if the same currency has been previously used. Therefore, conditional on other determinants of invoicing, the past invoicing share should predict current invoicing choices. This idea translates into the following econometric specification:

$$Peso_{f,opt} = \beta y_{f,t-1}^{Peso} + Controls_{f,opt} + FE_{fop} + FE_t + \nu_{f,opt}, \quad (17)$$

where $Peso_{f,opt}$ is a dummy variable equal to one when the transaction at the firm-origin-product(HS8)-time level is invoiced in Chilean peso and zero if in US dollar, $y_{f,t-1}^{Peso}$ is the lagged share of imports invoiced in Chilean peso at the firm level, and β is the coefficient of interest.

Including the lagged endogenous variable raises potential endogeneity concerns, as serially correlated unobserved factors may induce persistence in invoicing choices and upwardly bias the coefficient β . Following the literature, we include a comprehensive set of fixed effects to mitigate these concerns ([Bernard and Jensen, 2004](#); [Timoshenko, 2015](#); [Das et al., 2007](#)). Moreover, we introduce additional controls affecting invoicing choices, such as natural hedging and strategic complementarities, defined as in Equation (13).

Table 3: Sunk Cost Mechanism Validation

	OLS			IV
	(1)	(2)	(3)	(4)
	$Peso_{f,opt}$	$Peso_{f,opt}$	$Peso_{f,opt}$	$Peso_{f,opt}$
Lagged Peso Share	0.799*** (0.031)	0.094** (0.037)	0.097*** (0.037)	0.098*** (0.038)
Controls	No	No	Yes	Yes
Firm-Origin-HS8 & Year	No	Yes	Yes	Yes
Observations	42503	32556	31747	31747

Note: The Table reports coefficients from specification in Equation (17). The dependent variable equals 1 when the f_{opt} transaction is invoiced in Chilean peso and 0 otherwise. "Lagged Peso Share" refers to the lagged share of imports invoiced in local currency at the firm level. Controls include: i) a natural hedging measure defined as the share of exports invoiced in US dollars; ii) a strategic complementarities index defined as the average share of imports invoiced in US dollars among all competitors in a given sector(HS4)-origin pair as in Equation (14). IV columns refer to the case where the strategic complementarity index is instrumented with the import-weighted average US dollar exposure of all competitors in the sector(HS4)-origin pair as in Equation (15). The sample covers 1997–2017 from [Garcia-Marin et al. \(2019\)](#). Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3 shows that a higher lagged share of imports invoiced in local currency increases the probability of invoicing in Chilean pesos, consistent with the presence of sunk costs. The

OLS specification in column (1) indicates that a one-percentage-point increase in the lagged share of imports invoiced in local currency raises the probability of local-currency invoicing by 0.8 percentage points. The benchmark specification in column (4), which includes fixed effects and additional controls, shows that a one-percentage-point increase in the lagged share increases the probability of local-currency invoicing by 10 basis points. The smaller magnitude in the benchmark specification is consistent with an upward bias in the OLS estimates due to omitted fixed effects and controls.

Additional Robustness & Financial Hedging. Table 11 presents a robustness exercise that allows for heterogeneity in exchange-rate sensitivity by firms exposure to dollar invoicing. Column (1) uses the UIP deviation measure defined in Equation (12), but splits firms according to their natural hedging intensity, $NH_{ft}^{\$,H}$, defined as an indicator for having an export share invoiced in U.S. dollars above the sample mean (0.9). Column (2) adopts the same specification and definitions, but replaces UIP deviations with covered interest parity (CIP) deviations computed using one-year-ahead forward exchange rates between US dollar and Peso. Across both columns, deviations from parity increase the probability of peso invoicing, while this effect is significantly attenuated for firms with greater dollar exposure and for sectors with higher strategic complementarity. The CIP specification addresses the concern that firms may engage in financial hedging through forward markets, isolating the component of exchange-rate deviations that cannot be arbitrated away. The similarity of the results across UIP and CIP measures suggests that the results are not driven by financial hedging, but instead reflect firms operational and invoicing responses to currency risk.

Evidence Across Countries. We document similar UIP and invoicing dynamics around the GFC in a broader set of emerging market economies. We use a balanced panel of ten emerging-market currencies from January 2005 to December 2012, including economies geographically and economically comparable to Chile, such as Brazil, Mexico, Hungary, Israel, and Poland. We construct one-month realized UIP deviations against the U.S. dollar using daily spot and forward exchange rates.

Figure 18 in Appendix A.7 shows that UIP deviations spiked sharply during the GFC across all countries, exhibiting dynamics quantitatively similar to those observed for Chile. Using country-level invoicing data from Boz et al. (2020), the figure also shows that the share of imports invoiced in local currency increased on average—from about 1 percent to 4 percent—among these economies. Taken together, these patterns are consistent with the key mechanism in our theoretical framework operating more broadly: the global shortage of dollar funding during the GFC raised incentives to invoice imports in local currency across multiple emerging markets.

Bargaining in Invoicing. The invoicing pattern observed in the data implies that exporters located in the origin countries started to accept payments in Chilean peso. Arguably, Chilean

pesos are a less common currency compared to the ubiquitous US dollar. This suggests the existence of a relatively higher bargain power of Chilean importers when defining which currency to use to exchange goods internationally. We provide evidence that is coherent with the existence of a higher relative bargain power of Chilean importers with respect to foreign exporters.

First, the left panel of Figure 14 in Appendix A.5 shows a positive linear relationship between the cumulative change in the aggregate share of Chilean peso at the origin level and the Chilean export share in the origin's total exports in 2008. Intuitively, the graph suggests that, those origin countries whose total exports were relatively more dependent on Chile, they were also those countries whose share of transactions invoiced in local currency increased more. This is consistent with the idea that Chile was able to impose its currency to those counterparts that were more dependent from them.

Second, a model with strategic interactions in pricing among Chilean importers predicts an inverted U-shape relationship between exchange rate pass-through to import prices invoiced in local currency and importer's market share (Juarez, 2024). We confirm the existence of this pattern in the right panel of Figure 14 in Appendix A.5.

5 Aggregate Implications

In this section we quantitatively assess the macroeconomic implications of the large changes in invoicing shares after the GFC. We focus on the implications of invoicing choices for the dynamics of terms of trade and import inflation. We proceed as follow. We first estimate short-run and long-run price elasticities to exchange rate fluctuations accounting for the different invoicing currencies, building on existing empirical frameworks (Adler et al., 2020; Chen et al., 2022; De Gregorio et al., 2024). We then leverage the estimated elasticities and the evolution of invoicing shares to conduct accounting exercises (Auer et al., 2021). We find that the change in the invoicing shares after the GFC reduced import inflation by 5 percent while increasing terms of trade sensitivity to exchange rate fluctuations.

Price Elasticities to Exchange Rate We first estimate the exchange rate pass-through to border prices, both in the short-run and in the long-run, conditioning on the currency of invoicing. Following existing econometric frameworks, the benchmark specification is:

$$\begin{aligned} \Delta P_{f, opt}^k = & \sum_{l=0}^8 \beta_l^{LCP} \Delta e_{t-l}^{CLP/o} \times D_{f, opt}^{LCP} + \sum_{l=0}^8 \beta_l^{PCP} \Delta e_{t-l}^{CLP/o} \times D_{f, opt}^{PCP} + \\ & + \sum_{l=0}^8 \beta_l^{VCP} \Delta e_{t-l}^{CLP/\$} \times D_{f, opt}^{VCP} + FE_{f, opt} + \varepsilon_{f, opt}, \end{aligned} \quad (18)$$

where $\Delta x_t \equiv x_t - x_{t-1}$. Thus, $\Delta P_{f, opt}^k$ denotes the log change in unit values (expressed in Chilean pesos) of product p imported by firm f from origin o between quarters $t - 1$

and t , invoiced in currency k . An increase in $\Delta e_t^{CLP/o}$ corresponds to a depreciation of the Chilean peso relative to the currency of country o . The dummy variables D^{LCP} , D^{PCP} , and D^{DCP} indicate whether the transaction is invoiced in local, producer, or dominant currency, respectively. For a given invoicing scheme, short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^8 \beta_l$. We re-estimate Equation (18) using export unit values as dependent variable.

Table 4: Price Sensitivities to Exchange Rates

Invoicing	Imports		Exports	
	Short-run	Long-run	Short-run	Long-run
LCP	0.062 (0.068)	0.559 (0.260)	0.879 (0.029)	0.675 (0.088)
PCP	0.884 (0.016)	0.703 (0.055)	. (.)	. (.)
VCP	0.879 (0.014)	0.389 (0.037)	0.869 (0.023)	0.503 (0.045)

Note: The left (right) panel show the estimated coefficients from the specification in Equation (18) using import (export) prices as dependent variable. We use data from official Chilean customs running from 2007 to 2019 for imports and 2009 to 2019 for exports. We exclude the PCP case for export prices because of the absence of sufficient observations in the estimation. Values in parenthesis represents the standard errors clustered at the origin-time level. Short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^8 \beta_l$.

Table 4 reports the short-run and long-run exchange rate pass-through rates into import and export prices estimated from the main specification in Equation (18). Pass-through into import prices is not statistically different from zero when transactions are invoiced in local currency both in the short-run and long-run; pass-through is however almost complete when transactions are invoiced in dominant and partner currencies in the short-run but declines after two years. On the export side, pass-through rates are almost complete in the short-run when transactions are invoiced in dominant and local currencies.²⁵ In the long-run, the pass-through rate decreases to approximately 50-70 percent in both cases. These patterns are consistent with the idea that prices are sticky in the currency in which they are invoiced (Chen et al., 2022; Gopinath et al., 2010).

Effects on Terms of Trade and Import Inflation. In standard models of international economics (e.g. Mundell-Fleming), in which international prices are invoiced in the exporter's currency, exchange rates play a key role in external adjustment. However, the widespread use of US dollar invoicing, together with the differences in price elasticities documented above, impacts how the terms of trade and import inflation respond to exchange rate fluctuations.

Using accounting exercises as in Auer et al. (2021), we can define the sensitivity of terms of trade at horizon l to a depreciation of the Chilean peso in terms of invoicing shares and price elasticities as follows:

$$\Delta TOT_t^l = \Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PX,CLP} \times S^{X,CLP} + \beta_{\tau}^{PX,D} \times S^{X,D} + \beta_{\tau}^{PX,p} \times S^{X,p} \right] - \quad (19)$$

²⁵We do not report the pass-through rate into export prices conditional on being invoiced in producer currency (i.e. Chilean peso) because of a very restricted sample.

Table 5: ToT and IPI Sensitivity to Exchange Rates

Year	Import prices		Terms of trade	
	Short-run	Long-run	Short-run	Long-run
2007	0.876	0.464	-0.006	0.065
2019	0.832	0.457	0.033	0.073

Notes: The left (right) panel reports the response of the import price index (terms of trade) to a depreciation of the Chilean peso using Equation (20) (Equation (19)). We use the estimated coefficients from Equation (18) and reported in Table 4. Price elasticities not statistically different from zero are set to zero. Short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^8 \beta_l$. Invoicing shares are computed from official import (export) Chilean customs data from 2007 (2009) for the initial period and from 2019 for the last period.

$$\Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PM,CLP} \times S^{M,CLP} + \beta_{\tau}^{PM,D} \times S^{M,D} + \beta_{\tau}^{PM,p} \times S^{M,p} \right],$$

where $S^{X,z}$ ($S^{M,z}$) the share of exports (imports) invoiced in currency z , and $\beta_l^{j,z}$ is the elasticity of j to exchange rates when invoiced in currency z at horizon l , with j being import and export prices ($\{PX, PM\}$) and $z \in \{CLP, p, D\}$. Similarly, we can define the response of the import price index (IPI) to a depreciation of the Chilean peso. For each horizon l , we use the following formula:

$$\Delta IPI_t^l = \Delta e \left[\sum_{\tau=0}^l \beta_{\tau}^{PM,CLP} \times S^{M,CLP} + \beta_{\tau}^{PM,D} \times S^{M,D} + \beta_{\tau}^{PM,p} \times S^{M,p} \right]. \quad (20)$$

Table 5 shows that, relative to 2007, the import price index of Chile has become less sensitive to exchange rate fluctuations while the sensitivity of terms of trade has increased in 2019. The IPI sensitivity has decreased by approximately 5 percent in the short-run and 1.5 percent in the long-run. The sensitivity of the terms of trade to a one percent depreciation of the Chilean peso increased to 0.03 pp from zero in the short-run, and by 11 percent in the long run (0.073 vs 0.065). The key driver of these changes is the rise in the share of imports invoiced in local currency between 2007 and 2019 (Figure 1), which carries a lower pass-through to exchange rate fluctuations. Thus, abstracting from invoicing choices improperly informs on the effects of exchange rate fluctuations on key macroeconomic variables (Gopinath and Itskhoki, 2022; Barbiero, 2021).

Conclusion

Our analysis reveals how temporary shocks to dollar financing availability can weaken the US dollar invoicing incentives in international trade and boost alternative forms of invoicing, challenging the dominance of the US dollar. We document a gradual decline in the share of Chilean imports invoiced in US dollars over a decade after the spike in the cost of dollar financing during the Global Financial Crisis, accompanied by a corresponding increase in Chilean peso invoicing. Decomposition exercises show that surviving firms contribute approximately 80 percent of the observed shift. Our model explains firms invoicing decisions based on the relative costs of financing in domestic versus vehicle currencies, which can be proxied by UIP deviations, treating imported inputs as working capital. We show that temporary shocks to UIP deviations influence invoicing patterns, with sunk costs introducing hysteresis, permanently altering the convenience of using different currencies. We test the key mechanisms of our theory using firm-level data and various econometric techniques. Our findings support the key mechanisms, explaining half of the observed invoicing patterns in Chilean imports.

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A Appendix: Data & Robustness

A.1 Data

Table 6: Number of Invoicing Currencies within Firm-Product-Origin-Year Quartet

No. of Currencies	<i>fpot</i> Quartets	Share of Transactions (%)	Share of Value (%)
1	8384867	96.2	92.4
2	318226	3.65	7.07
3	16185	0.19	0.45
4	228	0.0026	0.029

Notes: The Table reports summary statistics on multi-currency use at the firm-product-origin-year (*fpot*) level. Specifically, we count the total number of different currencies that a firm uses to import a product (defined at the HS8 level) from a given origin in a given year. Data from Chilean Customs Agency from 2007 to 2019.

Table 7: Number of Invoicing Currencies within Firm-Product-Destination-Year Quartet

No. of Currencies	<i>fpot</i> Quartets	Share of Transactions (%)	Share of Value (%)
1	1225319	97.3	95.2
2	32726	2.60	4.73
3	1116	0.089	0.12

Notes: The Table reports summary statistics on multi-currency use at the firmproductoriginyear (*fpot*) level. Specifically, we count the total number of different currencies that a firm uses to export a product (defined at the HS8 level) from a given destination in a given year. Data from Chilean Customs Agency from 2009 to 2019.

Table 8: Summary Statistics for USD-CLP FX Forecasts

	StDev	Min	Max	N. Forecasters
2006	21.39	530	620	68
2007	32.53	470	620	129
2008	43.71	435	660	139
2009	59.94	430	725	157
2010	32.56	425	559	161
2011	29.17	420	540	164
2012	25.20	370	570	190
2013	34.36	370	570	190
Observations	1198			

Notes: The Table reports summary statistics of the one-year expected USD-CLP exchange rate. For each year we report the number of forecasters, the range of their forecast and the standard deviation of their forecasts. Data are sourced from Bloomberg.

A.2 Additional Decompositions

Table 9: Dynamic Olley-Pakes Decomposition with Entry and Exit: Imports at Origin Level

	USD	Peso	Euro	Others
-				
Aggregate Invoicing Share	-9.452	6.093	3.611	-0.251
Contribution Net Entry				
Total	-1.769	0.778	0.813	0.178
Firm Entry	-2.618	0.856	1.872	-0.110
Firm Exit	0.850	-0.078	-1.059	0.288
Contribution Incumbents				
Total	-7.683	5.315	2.798	-0.430
Within Firm	-4.560	6.367	-1.337	-0.470
Between Firm	-3.123	-1.052	4.134	0.041
Within Firm				
Total	-4.560	6.367	-1.337	-0.470
Net Entry Origin	0.634	0.080	-0.222	-0.491
Within Origin	-6.118	5.589	0.435	0.095
Between Origin	0.924	0.699	-1.549	-0.074

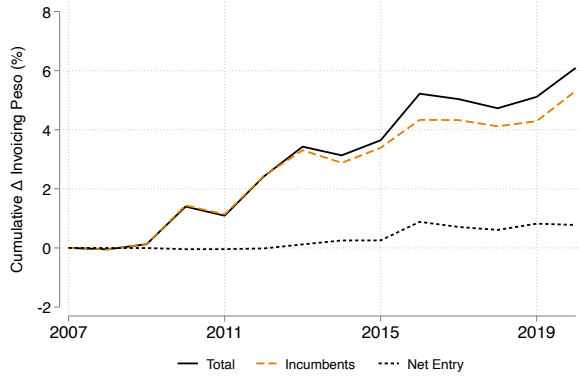
Notes: Table 9 reports the results from the dynamic OlleyPakes decomposition with entry and exit at the origin level for Chilean imports between 2007 and 2020. It corresponds to the decompositions in Equations (1), (2), and (3) where Equation (3) is properly adjust to consider only the origin level. One time period corresponds to one year.

Table 10: Olley-Pakes Decomposition of Exports Invoicing

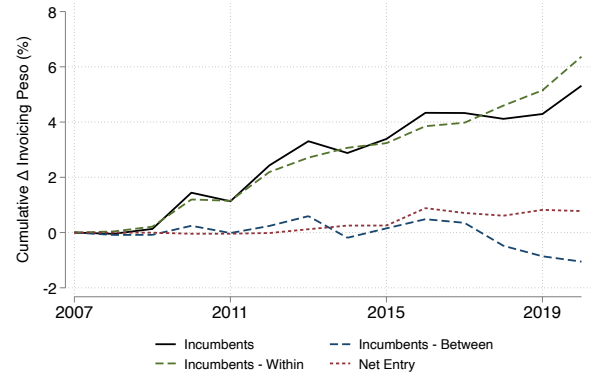
	USD	Peso	Euro	Others
Δ Aggregate Invoicing Share	0.488	0.009	-0.949	0.452
Net Entry	0.015	-0.000	-0.230	0.215
Firm Entry	-0.305	0.012	0.087	0.205
Firm Exit	0.319	-0.013	-0.317	0.010
Contribution Incumbents	0.473	0.009	-0.719	0.237
Within Firm	0.717	-0.326	-0.472	0.081
Between Firm	-0.244	0.335	-0.247	0.156
Within Firm				
Net Entry Destination-product	-1.551	2.107	-0.528	-0.028
Within Destination-product	0.932	-2.445	1.272	0.242
Between Destination-product	1.336	0.013	-1.215	-0.133

Notes: The Table reports the results from the dynamic OlleyPakes decomposition with entry and exit at the destinationproduct level for Chilean exports between 2009 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. We report the cumulative decomposition. Numbers are in percentage. Data are from the Chilean Customs Agency.

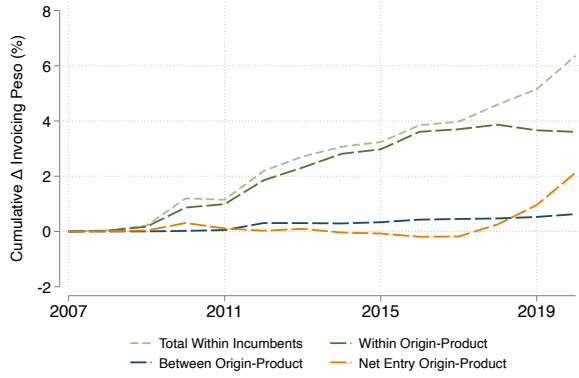
Figure 7: Decomposing Chilean Peso Invoicing Share over Time



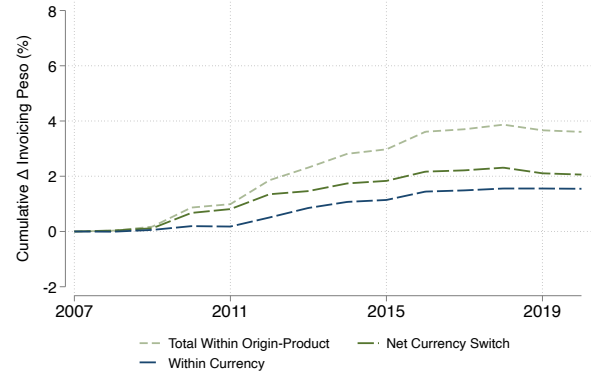
Panel (A): First Layer



Panel (B): First Layer



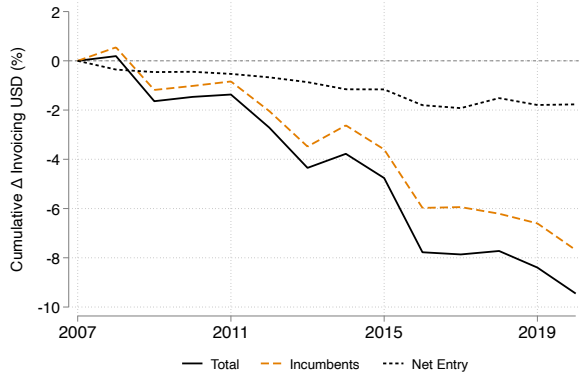
Panel (C): Second Layer



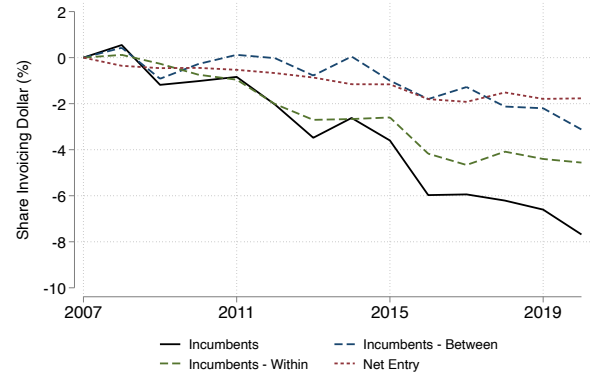
Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic OlleyPakes decomposition from Equations (??), (2), and (3) at the originproduct level for Chilean imports between 2007 and 2020. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in Chilean peso. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year. Data are from the Chilean Customs Agency.

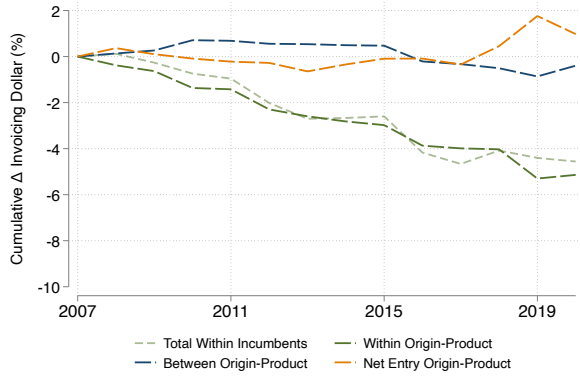
Figure 8: Decomposing US Dollar Invoicing Share over Time



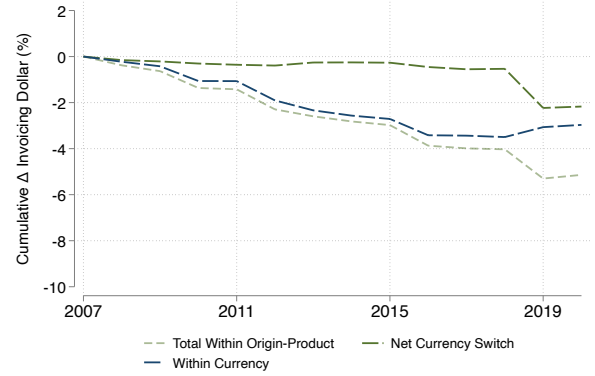
Panel (A): First Layer



Panel (B): First Layer



Panel (C): Second Layer

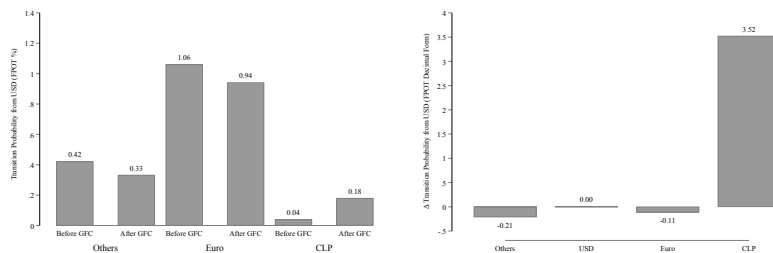


Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic OlleyPakes decomposition from Equations (??), (2), and (3) at the originproduct level for Chilean imports between 2007 and 2020. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in US dollar. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year. Data are from the Chilean Customs Agency.

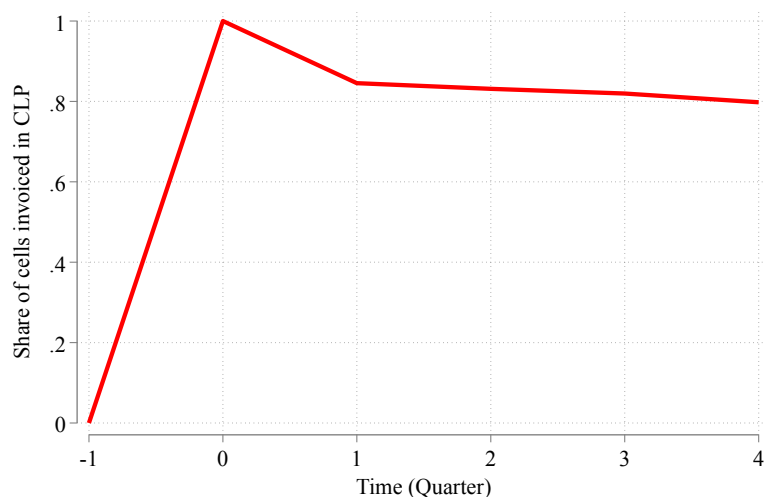
A.3 Additional Results on Invoicing and Switching Probability

Figure 9: Transition Probability from USD to CLP



Note: The left panel of the Figure reports the average transition probability from US dollar invoicing to Chilean peso invoicing over the two periods: before the GFC (2007-2009) and after (2010 onward). The right panel reports the percentage change in the transition probability between the two periods. Data are from the Chilean Customs Agency.

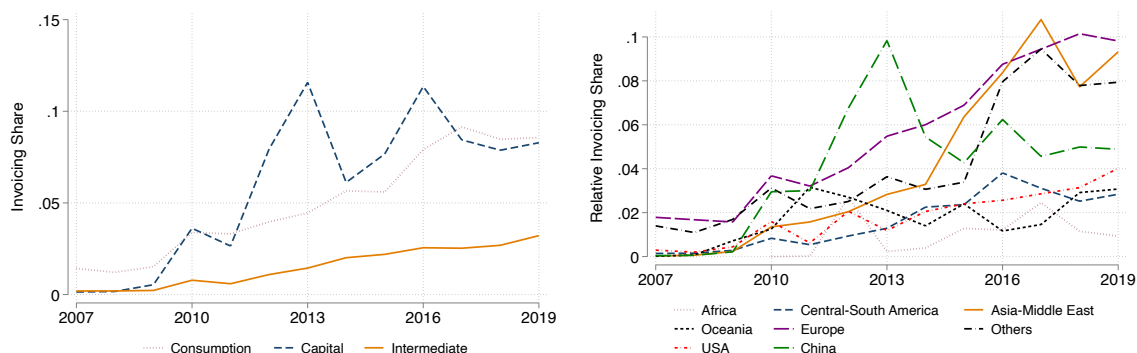
Figure 10: Persistence in Switching from US dollar to Chilean peso



Note: The Figure describes the invoicing currency in importerorigin-product im cells that switch for the first time from US dollars to Chilean peso in their imports to the four quarters after the switch. For each im cell, quarter zero corresponds to the first quarter in which the invoice currency is Chilean peso. We then compute the share of im cells invoicing in Chilean peso in each period. Data are from the Chilean Customs Agency.

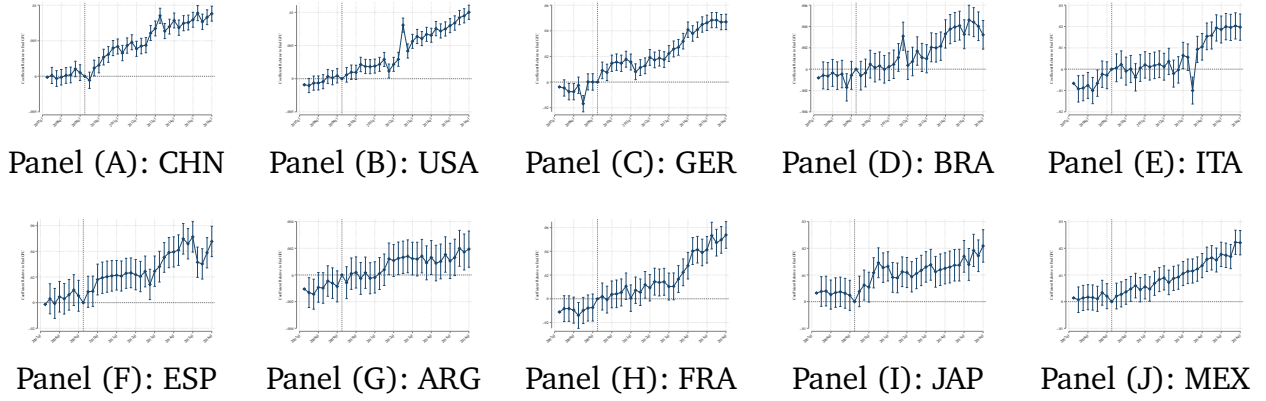
A.4 Additional Results: Product and Origin

Figure 11: Subgroup by Origin or Product Type - Chilean Peso Invoicing



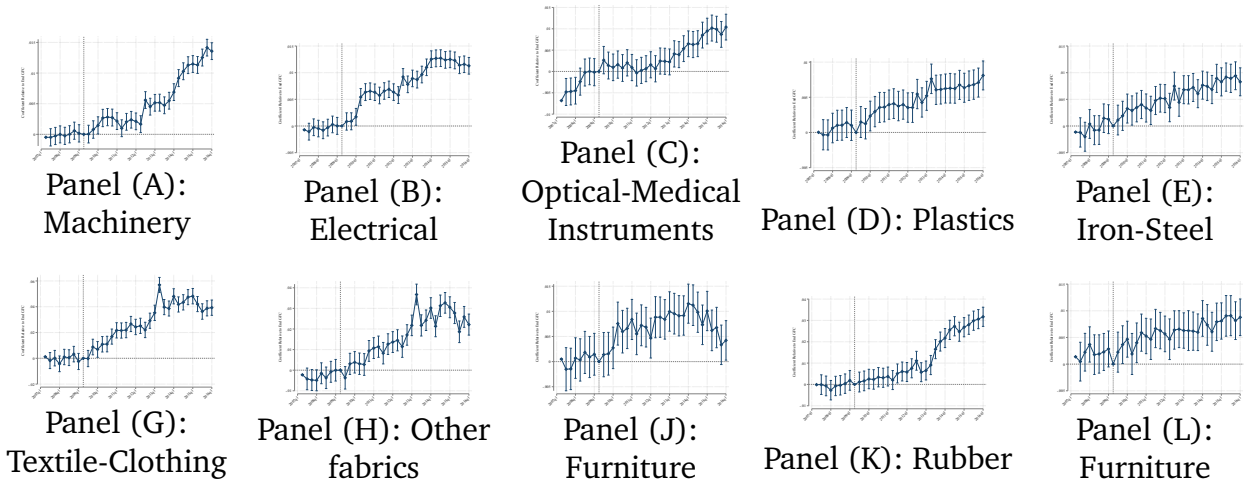
Notes: The left panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within three broad product categories: Consumption, capital, and investment goods as defined according to the BEC classification. The right panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within major trade partners: USA, Europe, China, Central-South America, Asia-Middle East, Africa and Others. Data are from the Chilean Customs Agency.

Figure 12: Effect of GFC on Probability of Invoicing in Chilean Peso: Main Origins



Notes: The Figure reports the results from Equation (4) estimated separately for each of top ten origin countries in the sample (ranked by number of observations at the *imt* level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2009Q2 (the end of the GFC). The vertical dashed line corresponds to the NBER recession period for the end of the GFC. Standard errors are clustered at the importerproductorigin level. Data are from the Chilean Customs Agency.

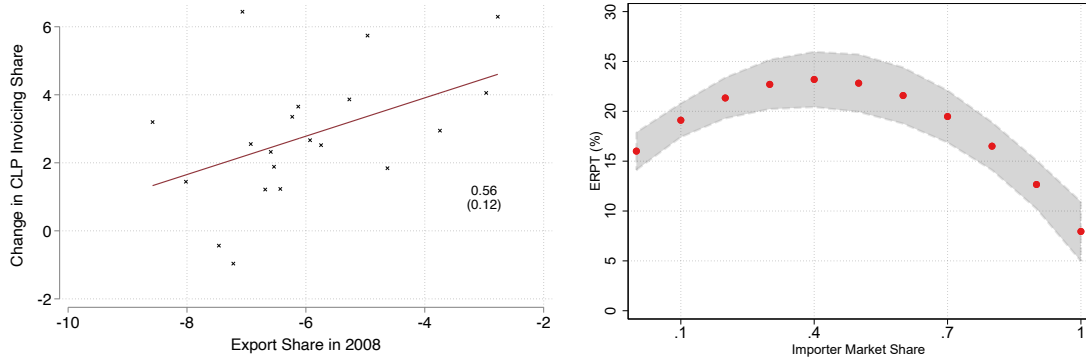
Figure 13: Effect of GFC on Probability of Invoicing in Chilean Peso: Main HS2



Notes: The Figure reports the results from Equation (4) estimated separately for each of top ten HS2 products in the sample (ranked by number of observations at the *imt* level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2009Q2 (the end of the GFC). The vertical dashed line corresponds to the NBER recession period for the end of the GFC. Standard errors are clustered at the importerproductorigin level. Data are from the Chilean Customs Agency.

A.5 Additional Results: Importers' Bargaining Power in Invoicing

Figure 14: Invoicing as Importers' Choice



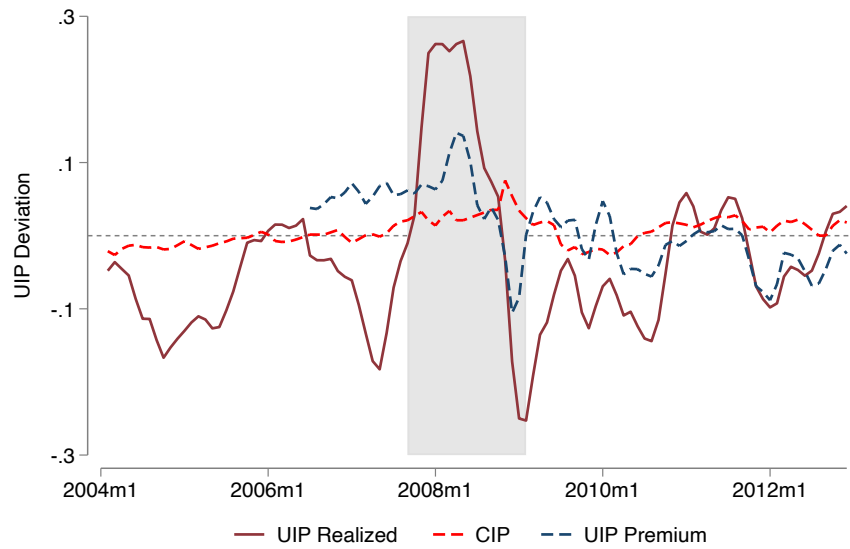
Notes: The left panel plots the relationship between the cumulative change in the aggregate share of CLP at the origin level and the Chilean export share in the origin's total exports in 2008 (in logs). We compute the cumulative change starting from 2008. We consider all horizons from 5 to 10 years, and plot the relationship after absorbing time fixed effects. The right panel plots the estimated quadratic relationship between the importer's market share and the pass-through rate of exchange rate fluctuations in import prices, estimated using the following specification:

$$\Delta \log p_{foit} = \beta_1 \Delta \log e_{ot} + \beta_2 S_{fst} + \beta_3 \Delta \log e_{ot} \times S_{fst} + \epsilon_{foit},$$

where S_{fst} is the market share of importer f in sector s (HS4-origin), and $\log p_{foit}$ is the unit import price in local currency for importer f , product i (HS8) from origin o .

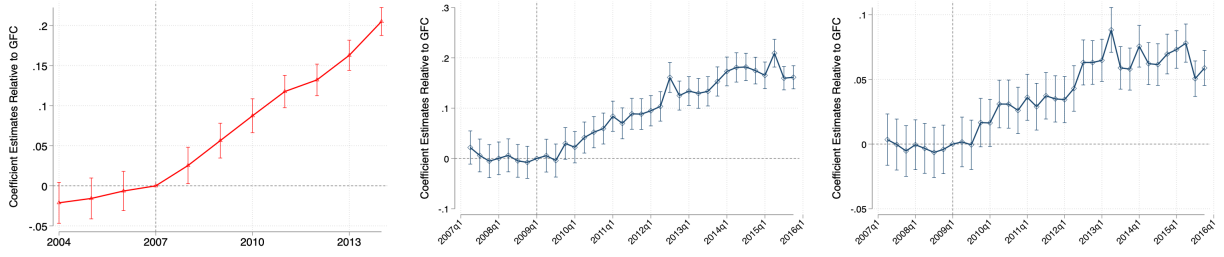
A.6 Additional Results: Model Validation

Figure 15: UIP Between the US dollar and Chilean peso



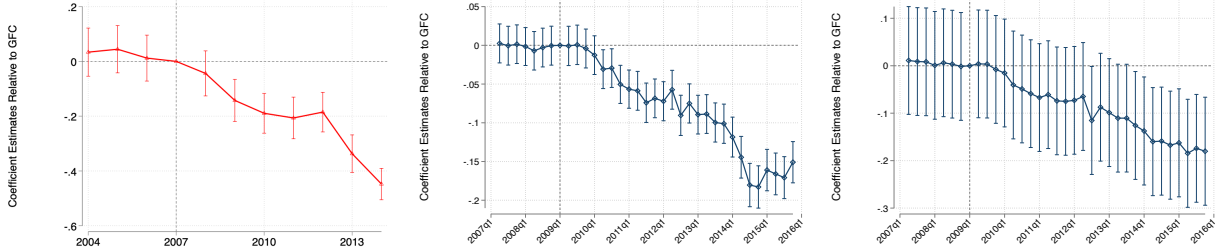
Note: The Figure plots the one-year ahead UIP deviation (dash blue), the realized one-year ahead UIP deviation (solid red), and the one-year CIP (dashed red) between the Chilean peso and the US dollar. A positive value indicates that borrowing in Chilean pesos is relatively cheaper than borrowing in US dollars. Interest rates are measured using commercial lending rates on short-term (less than one year) loans denominated in US dollars and Chilean pesos available from domestic banks. We construct the one-year-ahead expected exchange rate averaging the professional forecasters' expectations at the monthly frequency. We use non-deliverable forward (NDF) outright contracts for the U.S. dollar-peso from Bloomberg to construct measures of covered interest parity. Exchange rate forecasts are obtained from Bloomberg, available from 2006 onward. Table 8 reports summary statistics. Data are at the monthly frequency and smoothed using a 2-month rolling window.

Figure 16: Event Study - Exposure



Note: The Figure plots the coefficients β_s from the event study specification in Equation (16), capturing the differential increase in the probability of Chilean peso invoicing due to operational hedging measured by the share of imports denominated in US dollar (Proposition 2). In the left panel we use annual data from 2004 until 2014 from [Garcia-Marin et al. \(2019\)](#). Exposure is defined in 2007, and the shock is the change in UIP between 2007 and 2008. We include firm-product-origin fixed effects. In the middle panel we include origin-product-time fixed effects to the main specification in Equation (16). In the right panel we use discrete treatment by creating two groups in terms of operational hedging, above and below the 75th percentile of the distribution. We report 90 percent confidence intervals. Standard errors are clustered at firm-product-origin level.

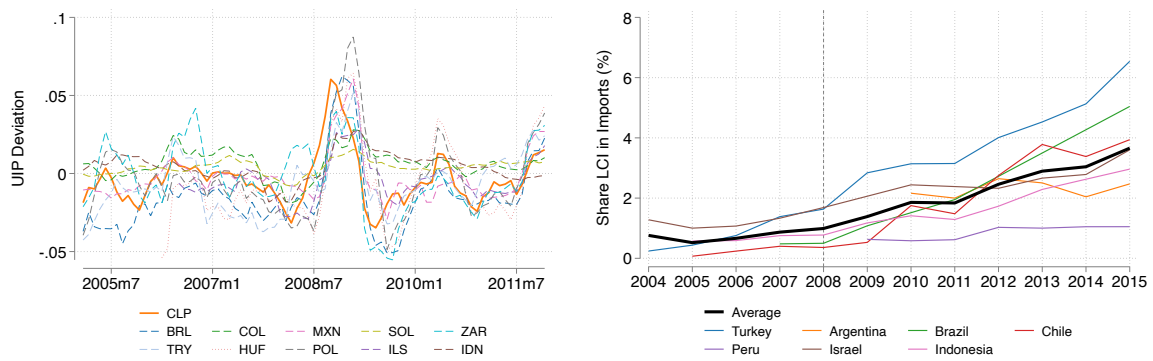
Figure 17: Event Study - Strategic Complementarity



Note: The Figure plots the coefficients β_s from the event study specification in Equation (16), capturing the differential increase in the probability of Chilean peso invoicing due to strategic complementarities in US dollar invoicing (Proposition 3). We use the strategic complementarity measure from Equation (14). In the left panel we use annual data from 2004 until 2014 from [Garcia-Marin et al. \(2019\)](#). Exposure is defined in 2007, and the shock is the change in UIP between 2007 and 2008. We include firm-product-origin and time fixed effects. In the middle panel we include origin-product fixed effects to the main specification in Equation (16). In the right panel we use discrete treatment by creating two groups in terms of strategic complementarity measure, above and below the 75th percentile of the distribution. We report 90 percent confidence intervals. Standard errors are clustered at firm-product-origin level.

A.7 Additional Results: Cross-Country Evidence of Invoicing Currency

Figure 18: Cross-Country Evidence of Invoicing Currency



Notes: The left panel reports the 1-month realized UIP deviation relative to the US for the following set of EMs: Turkey, Brazil, Colombia, Mexico, Chile, Peru, Israel, South Africa, Hungary, Poland, and Indonesia. Interest rate differential are computed using forward exchange rates, under the assumption that CPI holds. Data are from Datastream and Thomson Reuters for the period from 2005 until 2012. Series are smoothed using a six-month rolling window. The right panel reports the share of imports invoiced in local currency for the following set of EMs: Turkey, Argentina, Brazil, Chile, Peru, Israel, and Indonesia. The solid black line represents the average. Data are from [Boz et al. \(2020\)](#) for the period from 2004 to 2014. The dataset is available at <https://www.imf.org/~media/Files/Publications/WP/2020/Datasets/wp20126.ashx>.

A.8 Additional Results: Robustness & Financial Hedging

Table 11: Cross-Sectional Analysis: Robustness

	UIP	CIP
	(1)	(2)
	$Pr(Peso)_{f_{opt}}$	$Pr(Peso)_{f_{opt}}$
$UIP_t^{\$,PESO}$	0.219*** (0.014)	
$SC_{kt}^{\$} \times UIP_t^{\$,PESO}$	-0.161*** (0.012)	
$NH_{ft}^{\$,H} \times UIP_t^{\$,PESO}$	-0.062*** (0.007)	
$SC_{kt}^{\$}$	-0.199*** (0.005)	-0.192*** (0.005)
$NH_{ft}^{\$,H}$	-0.007*** (0.001)	-0.006*** (0.001)
$CIP_t^{\$,PESO}$		0.635*** (0.048)
$SC_{kt}^{\$} \times CIP_t^{\$,PESO}$		-0.572*** (0.046)
$NH_{ft}^{\$,H} \times CIP_t^{\$,PESO}$		-0.086*** (0.023)
Firm-Origin-HS8	Yes	Yes
Observations	965928	965928

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B Appendix: Theory

B.1 Firm Invoicing Choice.

Start from the firm maximization problem displayed in Equation (9). For any given η_i , the optimal domestic price $p_i^{d*} = \frac{\theta}{\theta-1} E[C_i(s^v, \eta_i)]$. A firm invoices working capital x_i in vehicle currency v ($\eta_i^* = 0$) if and only if expected profit for doing so is higher or equal to the expected profit in the complementary case when working capital is invoiced in domestic currency ($\eta_i^* = 1$). That is:

$$\eta_i^* = 0 \quad \text{iff} \\ p_i^d (p_i^d)^{-\theta} - \mathbb{E}[C_i(s^v, 0)] (p_i^d)^{-\theta} - F_i^v \geq p_i^d (p_i^d)^{-\theta} - \mathbb{E}[C_i(s^v, 1)] (p_i^d)^{-\theta} - F_i^d$$

Plug in the optimal price expression for the respective cases and obtain:

$$\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \right)^{-\theta} - \mathbb{E}[C_i(s^v, 0)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 0)] \right)^{-\theta} - F_i^v \geq \\ \frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \right)^{-\theta} - \mathbb{E}[C_i(s^v, 1)] \left(\frac{\theta}{\theta-1} \mathbb{E}[C_i(s^v, 1)] \right)^{-\theta} - F_i^d$$

Which simplifies to:

$$\frac{(\theta-1)^{\theta-1}}{\theta^\theta} [\mathbb{E}[C_i(s^v, 0)]^{1-\theta} - \mathbb{E}[C_i(s^v, 1)]^{1-\theta}] \geq F_i^v - F_i^d$$

Then, consider the expression for the expected marginal cost of the firm for the two different cases:

$$\mathbb{E}[C_i(s^v, 0)] = w^{1-\alpha_i} \mathbb{E}[(1+i_v)\rho_v s_v]^{\alpha_i} \\ \mathbb{E}[C_i(s^v, 1)] = w^{1-\alpha_i} [(1+i_d)\rho_d]^{\alpha_i}$$

Re-write the expression above as:

$$\mathbb{E}[(1+i_v)\rho_v s_v]^{\alpha_i(1-\theta)} - [(1+i_d)\rho_d]^{\alpha_i(1-\theta)} \geq \tau_i \times (F_i^v - F_i^d) \\ \tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}$$

which is the expression in the main text. Note that $\frac{\theta^\theta}{(\theta-1)^{\theta-1}} > 1$ for $\theta > 1$ and it is strictly increasing in θ .

Then, consider the simplified case in which $F_i^v = F_i^d$ and $s^v \sim \text{Log-Normal}(\mu, \Sigma)$. Then,

$$((1+i_v)\rho_v)^{\alpha_i(1-\theta)} \mathbb{E}[s_v^{\alpha_i}]^{1-\theta} \geq [(1+i_d)\rho_d]^{\alpha_i(1-\theta)}$$

Then, i) elevate both sides to the power of $1/(1-\theta)$, ii) take logs on both sides, iii) use the

result that $\mathbb{E}[s_v^{\alpha_i}] = \exp(\alpha_i \mu + 0.5 \times \alpha_i^2 \Sigma)$, and obtain the expression in the text.

B.2 Proof of Proposition 1.

Manipulating Equation (10) we can write:

$$A(1 + i_v)^{\alpha_i(1-\theta)} - B > C \quad \text{with } A \equiv \mathbb{E}[(\rho_v s_v)^{\alpha_i}]^{1-\theta}; B \equiv [\rho_d(1 + i_d)]^{\alpha_i(1-\theta)}; C \equiv \tau_i (F_i^v - F_i^d). \quad (21)$$

Changing i_v scales only the first term in the left-hand side. Given that $\alpha_i > 0$ and $\theta > 1$, increasing i_v lowers the left-hand side and therefore reduces the chance the inequality holds and the probability of invoicing in vehicle currency.

B.3 Proof of Proposition 2.

Further manipulating Equation (21), we obtain the threshold i_v^* that makes importers indifferent between vehicle and local currency invoicing:

$$i_v^* = \exp \left(\frac{\log(B + C)}{\alpha_i(1 - \theta)} - \log \rho_v - \mu - 0.5\alpha_i \Sigma \right) - 1, \quad (22)$$

where we used the fact that s_v is distributed according to a log-normal with mean μ and variance Σ . We prove Proposition 2 by showing that the threshold i_v^* is decreasing in α_i . Increasing i_v makes the condition $\eta_i^* = 0$, i.e. vehicle currency invoicing, less likely. The threshold determines the cutoff, so how α_i affects the sensitivity of the 0-1 outcome to changes in i_v is captured by how the threshold i_v^* moves with α_i . It can be shown that:

$$\frac{\partial i_v^*}{\partial \alpha} \equiv (1 + i_v^*) \left(\frac{\log(B + C)}{\alpha_i^2(\theta - 1)} - 0.5\Sigma \right).$$

Proposition 2 predicts that $\frac{\partial i_v^*}{\partial \alpha} < 0$, i.e. higher α_i lowers the threshold, reducing the region in which $\eta_i^* = 0$ and making the probability of vehicle currency lower when i_v rises. The sign of the derivative above is negative for sufficiently high variance of the exchange rate Σ . This is in line with the key intuition. Firms maintain a constant markup over marginal costs, so vehicle-currency invoicing implies that ex-post marginal costs depend on the realized exchange rate, potentially generating costly deviations from the optimal markup. Thus, firms choose vehicle-currency invoicing only when its financing cost advantage outweighs the risk of markup misalignment, which ultimately depends on the volatility Σ .

B.4 Proof of Proposition 3.

We can prove Proposition 3 by applying the same reasoning as above and differentiating Equation (22) with respect to the fixed cost F_v :

$$\frac{\partial i_v^*}{\partial F_v} \equiv \frac{\tau_i}{\alpha_i(1-\theta)} \frac{(1+i_v^*)}{B+C} < 0.$$

It follows that lower F_v , or equivalently stronger strategic complementarities in vehicle currency invoicing, loosen the inequality making $\eta_i^* = 0$ more likely.